

```

root_path = fileparts(mfilename("fullpath"));
data_path = fullfile(root_path, 'sym4D_cutsAndFits');

if ~exist('cuts2fit400', 'var')
    ld = load(fullfile(data_path, 'multicuts_fit_dataGH_ei400meV.mat'));
    cuts2fit400 = ld.cuts2fit400;
end
used_ds = containers.Map()

```

```
used_ds =
```

```
Map with properties:
```

```

    Count: 0
   KeyType: char
  ValueType: any

```

```

sel_info = struct('cuts', [], 'en_range', [], 'fname', '', 'bg_par', []);
%
sel_info.cuts = cuts2fit400.cuts_dir001FFfit(1);
sel_info.en_range = 50:10:160;
sel_info.fname='EnFit_Ei400ref001off110_2D_constFFCut';
sel_info.bg_par= {};
used_ds('Ei400Br110dir001')=sel_info;
%
sel_info.cuts = cuts2fit400.cuts_dir001FFfit(2);
sel_info.en_range = 50:10:200;
sel_info.fname='EnFit_Ei400ref001off200_2D_constFFCut';
sel_info.bg_par= {};
used_ds('Ei400Br200dir001')=sel_info;

```

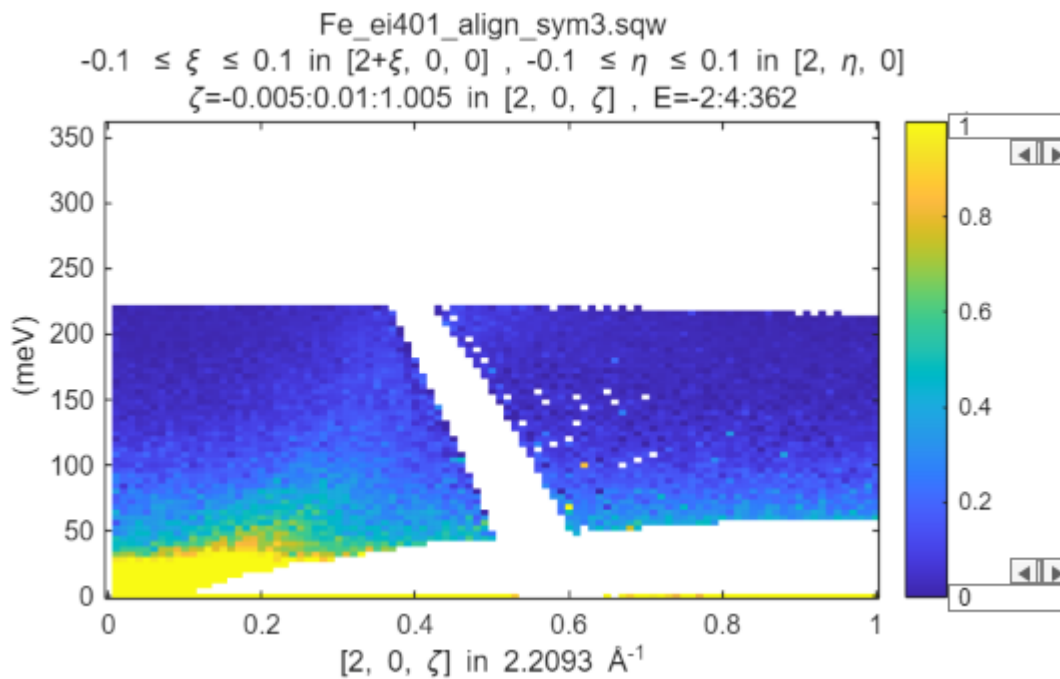
```

%cuts_list = cuts2fit400.cutsS_list; % all available cuts
%cuts_list = {cuts2fit400.cut_b200_200sel}; % cuts in low ff direction around
<200>
%cuts_list = cuts2fit400.cuts_dir001FFfit(1);
select= 'Ei400Br200dir001';
sel_info = used_ds(select);

cuts_list = sel_info.cuts;

mi = maps_instrument(400,600, 'S');
sample=IX_sample(true,[1,0,0],[0,1,0], 'cuboid',[0.04,0.03,0.02]);
for i=1:numel(cuts_list)
    cuts_list{i} = cuts_list{i}.set_instrument(mi);
    cuts_list{i} = cuts_list{i}.set_sample(sample);
    plot(cuts_list{i}); lz 0 1; keep_figure;
end

```

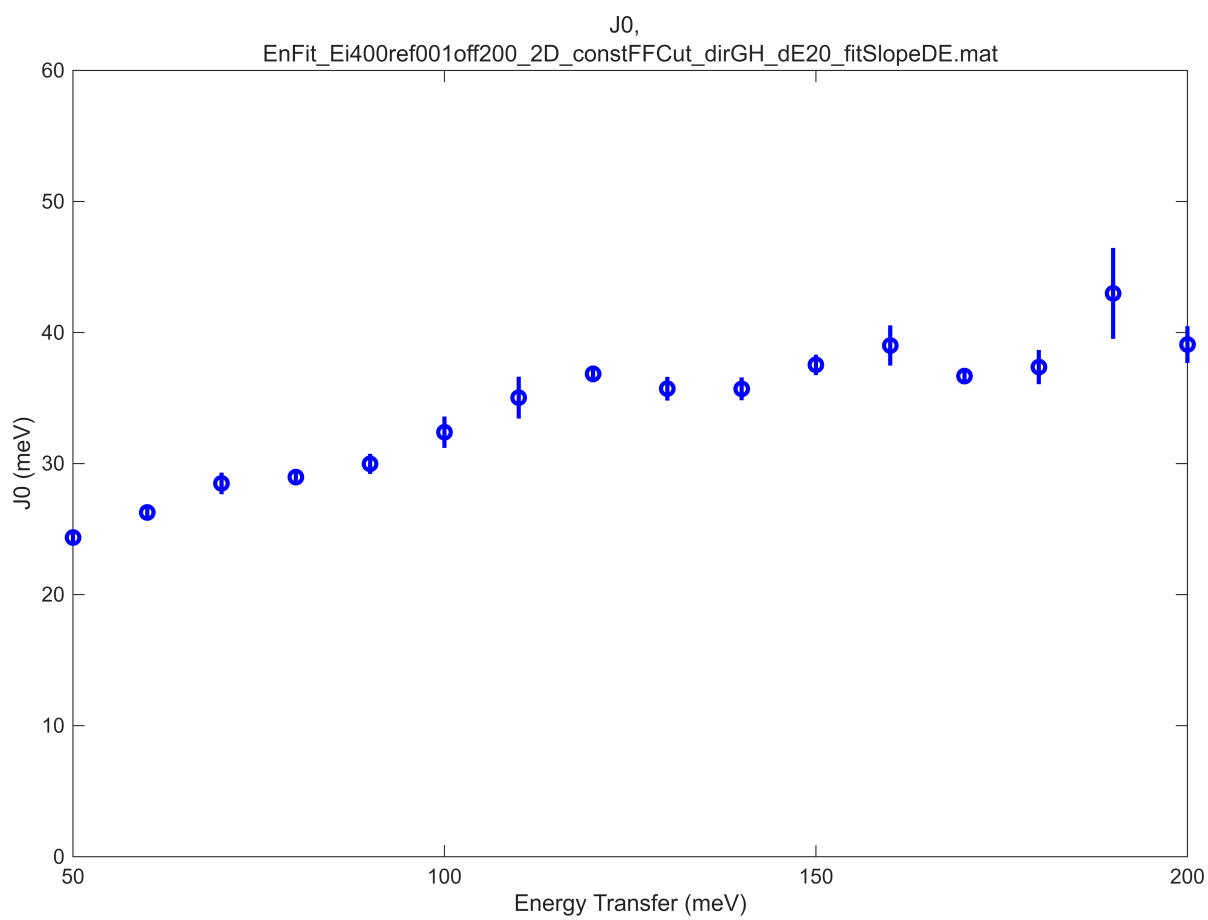
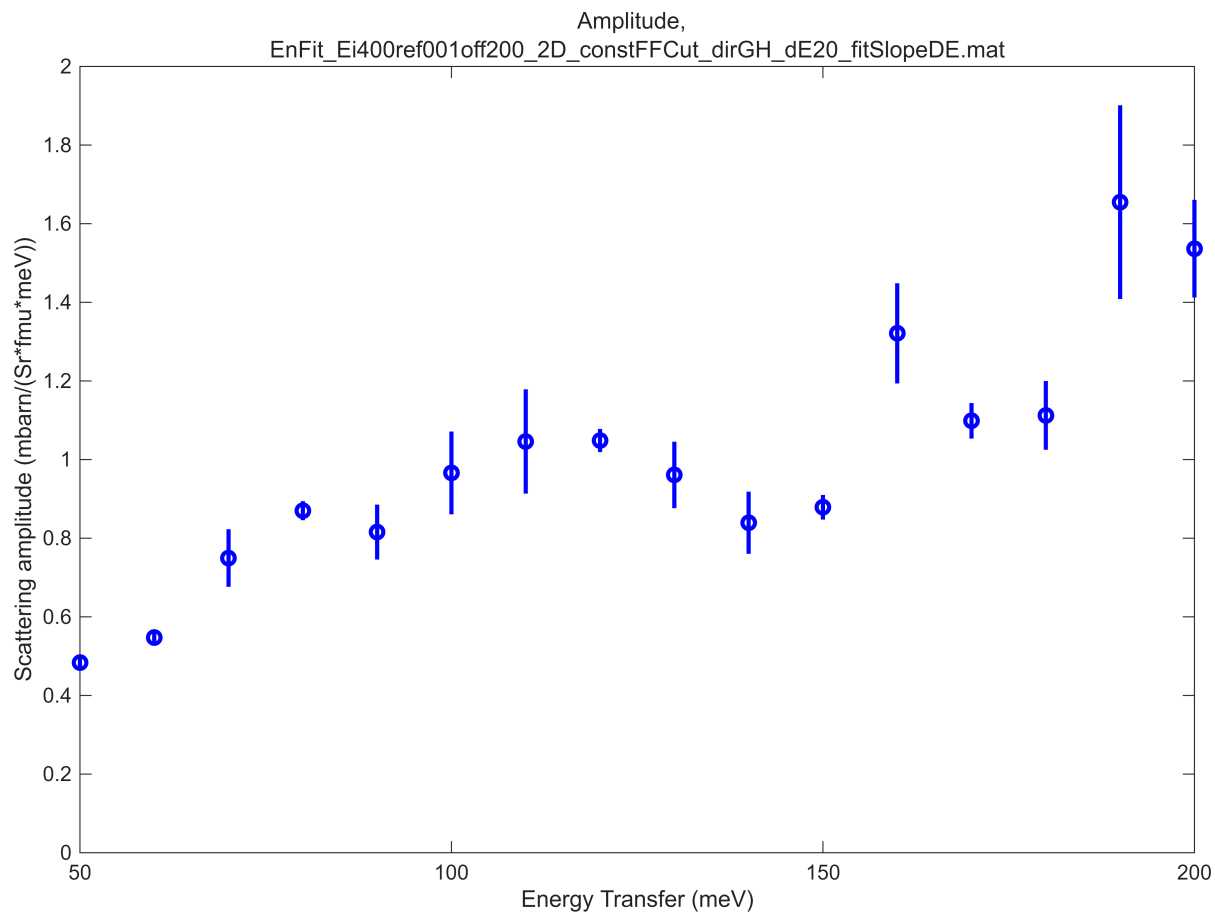


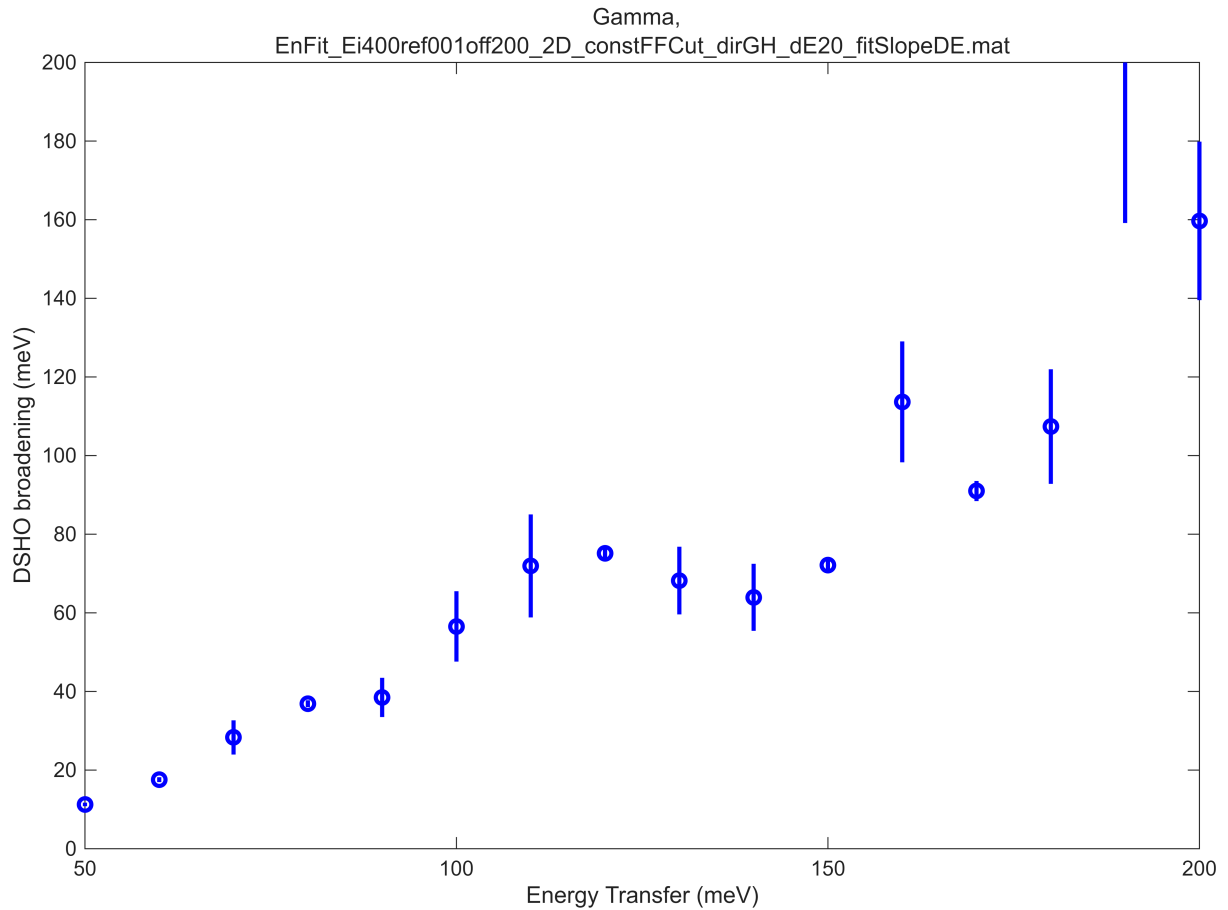
```
dir_name = "GH";

dE_step = 4; %original energy transfer step data were binned to. No point in
going finer
half_dE = 5; % half width of data binning
%cut_en = 50:10:260;
%cut_en = 50:10:160; %cuts2fit400.cuts_dir001FFfit(1);
cut_en = sel_info.en_range; %cuts2fit400.cuts_dir001FFfit(2);

%cl= {cuts_list{3}};
%cl = {w2_test_cut};
% if isfield(cuts2fit400,'bg_par_010off000')
%     if numel(cuts_list) == 3
%         bg_par =
{cuts2fit400.bg_par_010off000.p,cuts2fit400.bg_par_010off100.p,cuts2fit400.bg_par
_010off200.p};
%     else
%         bg_par = {cuts2fit400.bg_par_010off100.p};
%     end
%     cuts_list{end+1} = bg_par;
% end
bg_par = sel_info.bg_par;
cuts_list{end+1} = bg_par;
fbase_name = sel_info.fname;
```

```
[fit_res_400rec,res_name] = fit_multicuts_along_direction(...
    cuts_list,fbase_name,dir_name,cut_en,dE_step,half_dE);
```





**** En = 50±5

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested

*** Step 1 of 1; Read data for 69914 pixels -- processing data... -----> included 43501 pixels

Number of points in input object: 69914

Fraction of object selected: 100.00% (= 69914 points) ; Access speed: 1.43GB/sec

Fraction of object retained: 62.22% (= 43501 points)

**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 12.5%; Transf. time frac.: 80.0%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept

*** Step 1 of 1; Read data for 69914 pixels -- processing data... -----> retained 43501 pixels

Number of points in input object: 69914

Fraction of object selected: 100.00% (= 69914 points) ; Access speed: 1.62GB/sec

Fraction of object retained: 62.22% (= 43501 points)

**** Total time in cut(sqw) : 0.0sec Including:

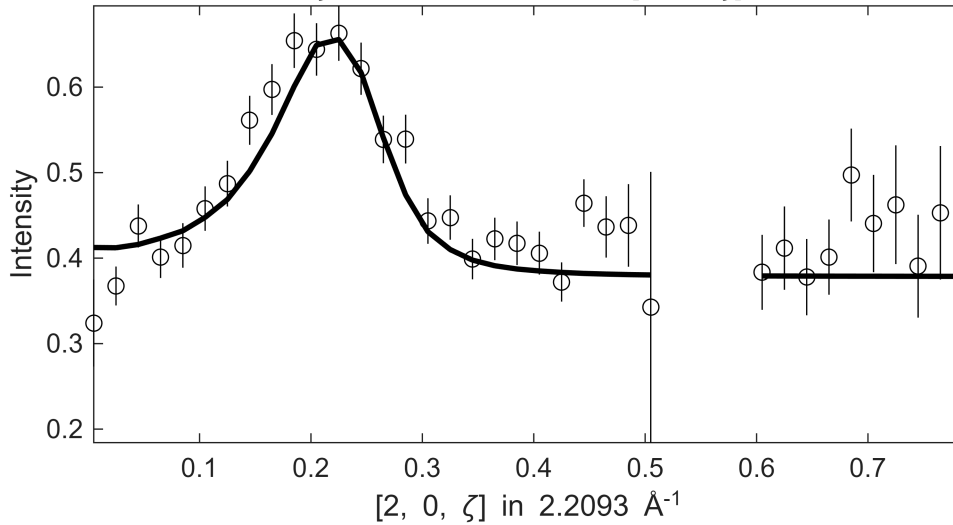
Data access time fraction: 10.3%; Transf. time frac.: 59.0%; Resulting pix preparation: 27.2%

Fe_ei401_align_sym3.sqw

S=0.59±0.081; J0= 24± 0.7; gamma=12.6067±3.34144

-0.1 ≤ ξ ≤ 0.1 in [2+ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, η, 0] , 45 ≤ E ≤ 55

ζ=-0.005:0.02:1.015 in [2, 0, ζ]



*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
 *** Step 1 of 1; Read data for 69914 pixels -- processing data... -----> retained 43501 pixels

Number of points in input object: 69914

Fraction of object selected: 100.00% (= 69914 points) ; Access speed: 1.94GB/sec

Fraction of object retained: 62.22% (= 43501 points)

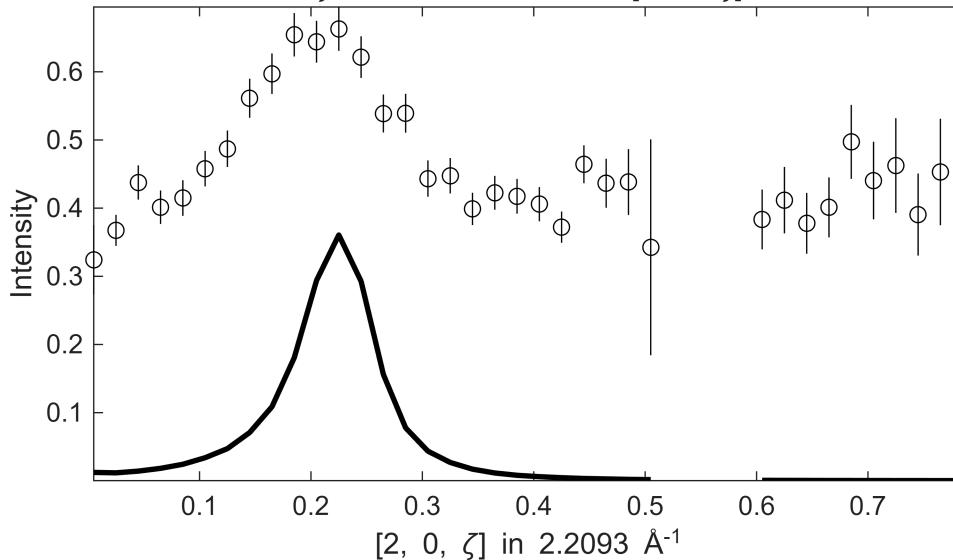
**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 17.1%; Transf. time frac.: 66.4%; Resulting pix preparation: 16.5%

Fe_ei401_align_sym3.sqw

-0.1 ≤ ξ ≤ 0.1 in [2+ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, η, 0] , 45 ≤ E ≤ 55

ζ=-0.005:0.02:1.015 in [2, 0, ζ]



**** En = 60±5

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested

*** Step 1 of 1; Read data for 93946 pixels -- processing data... -----> included 63157 pixels

Number of points in input object: 93946

Fraction of object selected: 100.00% (= 93946 points) ; Access speed: 2.20GB/sec

Fraction of object retained: 67.23% (= 63157 points)

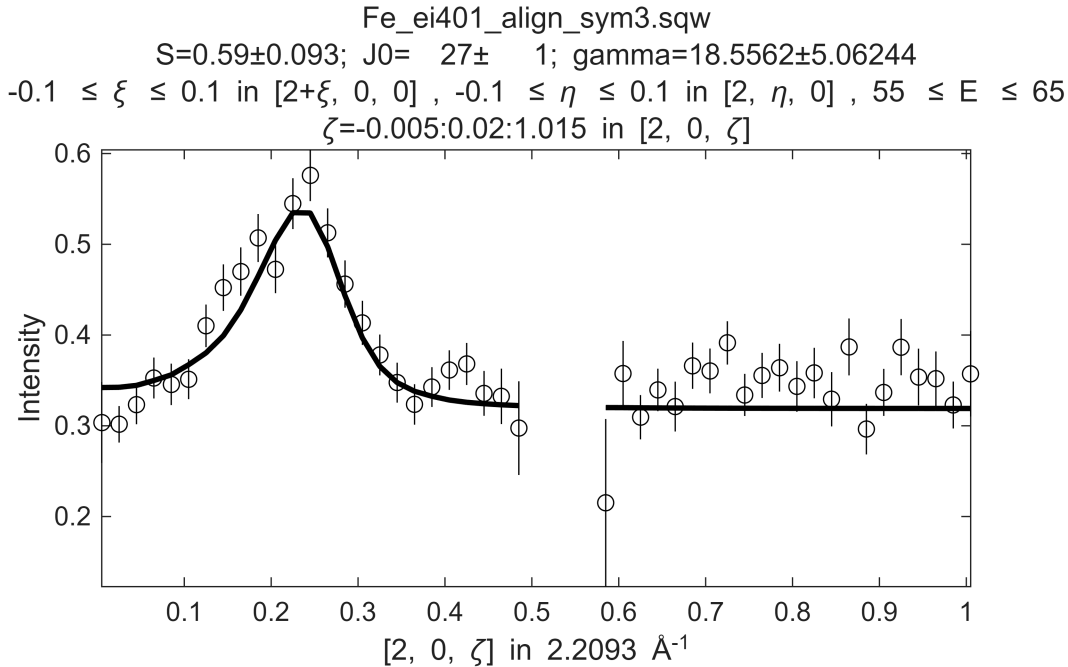
**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 21.1%; Transf. time frac.: 74.9%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 93946 pixels -- processing data... -----> retained 63157 pixels

Number of points in input object: 93946
Fraction of object selected: 100.00% (= 93946 points) ; Access speed: 1.98GB/sec
Fraction of object retained: 67.23% (= 63157 points)

**** Total time in cut(sqw) : 0.0sec Including:
Data access time fraction: 18.4%; Transf. time frac.: 65.1%; Resulting pix preparation: 16.5%

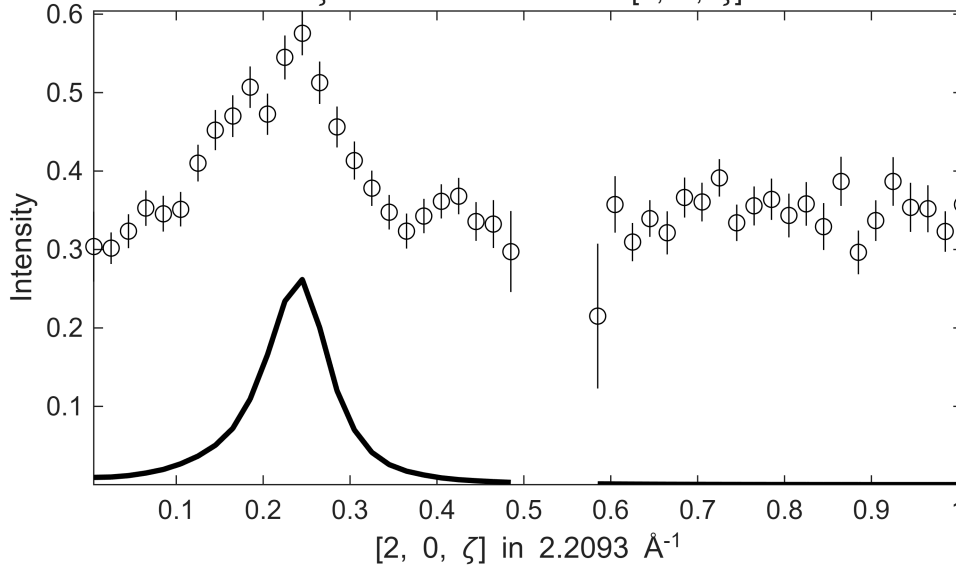


*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 93946 pixels -- processing data... -----> retained 63157 pixels

Number of points in input object: 93946
Fraction of object selected: 100.00% (= 93946 points) ; Access speed: 2.00GB/sec
Fraction of object retained: 67.23% (= 63157 points)

**** Total time in cut(sqw) : 0.0sec Including:
Data access time fraction: 17.2%; Transf. time frac.: 65.7%; Resulting pix preparation: 17.0%

Fe_ei401_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[2+\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, \eta, 0]$, $55 \leq E \leq 65$
 $\zeta = -0.005:0.02:1.015$ in $[2, 0, \zeta]$



**** En = 70±5

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested

*** Step 1 of 1; Read data for 102187 pixels -- processing data... -----> included 69052 pixels

Number of points in input object: 102187

Fraction of object selected: 100.00% (= 102187 points) ; Access speed: 2.04GB/sec

Fraction of object retained: 67.57% (= 69052 points)

**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 30.5%; Transf. time frac.: 62.7%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept

*** Step 1 of 1; Read data for 102187 pixels -- processing data... -----> retained 69052 pixels

Number of points in input object: 102187

Fraction of object selected: 100.00% (= 102187 points) ; Access speed: 1.88GB/sec

Fraction of object retained: 67.57% (= 69052 points)

**** Total time in cut(sqw) : 0.0sec Including:

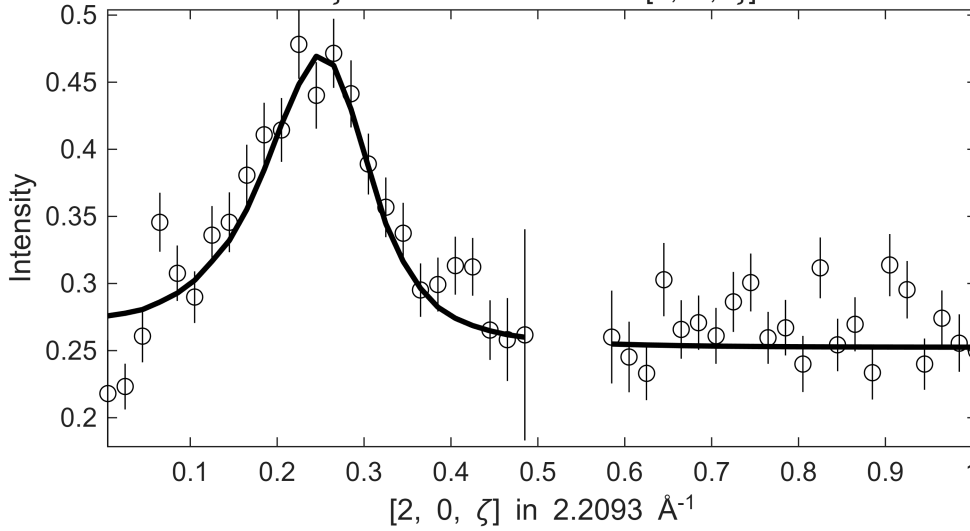
Data access time fraction: 19.4%; Transf. time frac.: 60.5%; Resulting pix preparation: 20.1%

Fe_ei401_align_sym3.sqw

S=0.85±0.12; J0= 29± 1.2; gamma=31.4865±6.77117

-0.1 ≤ ξ ≤ 0.1 in [2+ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, η, 0] , 65 ≤ E ≤ 75

ζ=-0.005:0.02:1.015 in [2, 0, ζ]



*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
 *** Step 1 of 1; Read data for 102187 pixels -- processing data... -----> retained 69052 pixels

Number of points in input object: 102187

Fraction of object selected: 100.00% (= 102187 points) ; Access speed: 2.02GB/sec

Fraction of object retained: 67.57% (= 69052 points)

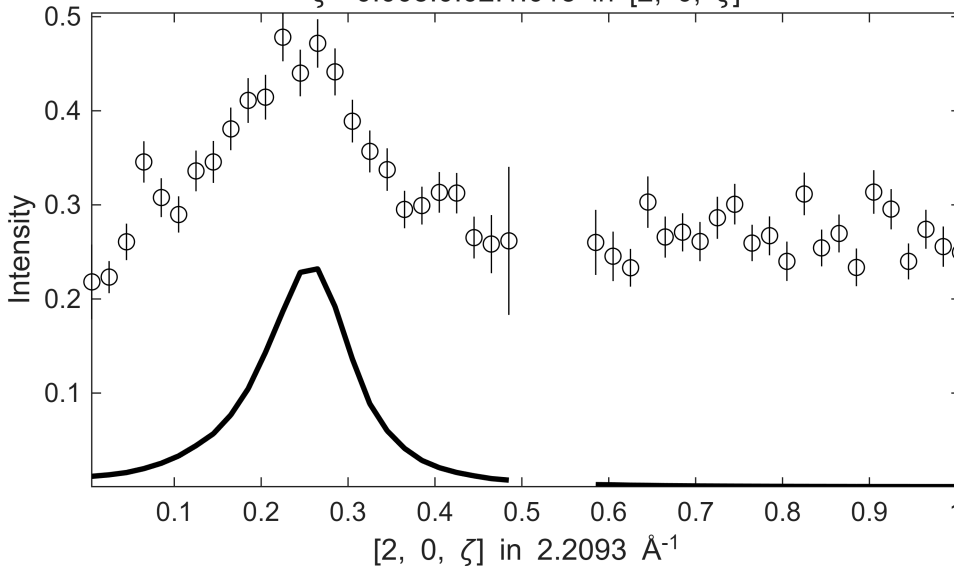
**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 18.7%; Transf. time frac.: 64.7%; Resulting pix preparation: 16.6%

Fe_ei401_align_sym3.sqw

-0.1 ≤ ξ ≤ 0.1 in [2+ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, η, 0] , 65 ≤ E ≤ 75

ζ=-0.005:0.02:1.015 in [2, 0, ζ]



**** En = 80±5

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested

*** Step 1 of 1; Read data for 99690 pixels -- processing data... -----> included 65691 pixels

Number of points in input object: 99690

Fraction of object selected: 100.00% (= 99690 points) ; Access speed: 2.10GB/sec

Fraction of object retained: 65.90% (= 65691 points)

**** Total time in cut(sqw) : 0.0sec Including:

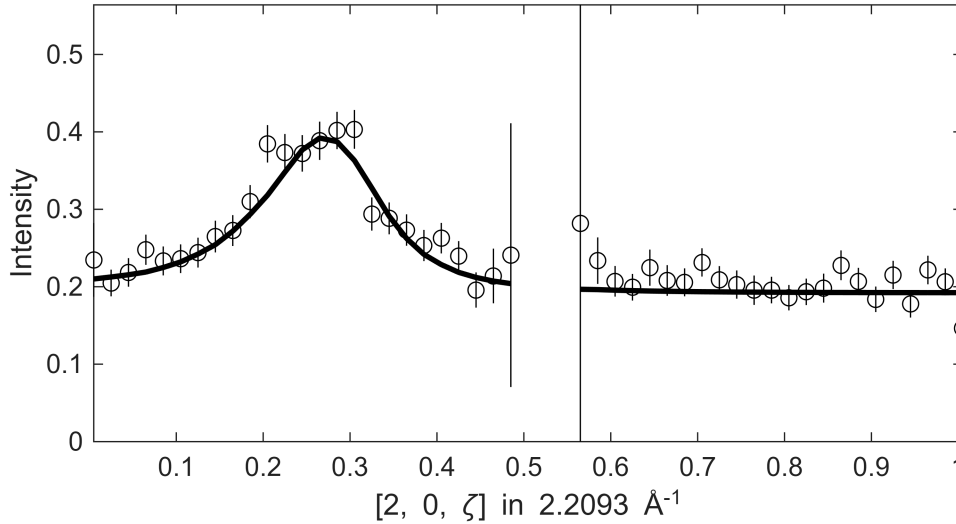
Data access time fraction: 20.0%; Transf. time frac.: 75.6%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 99690 pixels -- processing data... -----> retained 65691 pixels

Number of points in input object: 99690
Fraction of object selected: 100.00% (= 99690 points) ; Access speed: 1.97GB/sec
Fraction of object retained: 65.90% (= 65691 points)

**** Total time in cut(sqw) : 0.0sec Including:
Data access time fraction: 16.1%; Transf. time frac.: 64.7%; Resulting pix preparation: 19.1%

Fe_ei401_align_sym3.sqw
S=0.95±0.12; J0= 29± 1.2; gamma=39.8148±7.79478
-0.1 ≤ ξ ≤ 0.1 in [2+ ξ , 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, η , 0] , 75 ≤ E ≤ 85
 ζ =-0.005:0.02:1.015 in [2, 0, ζ]

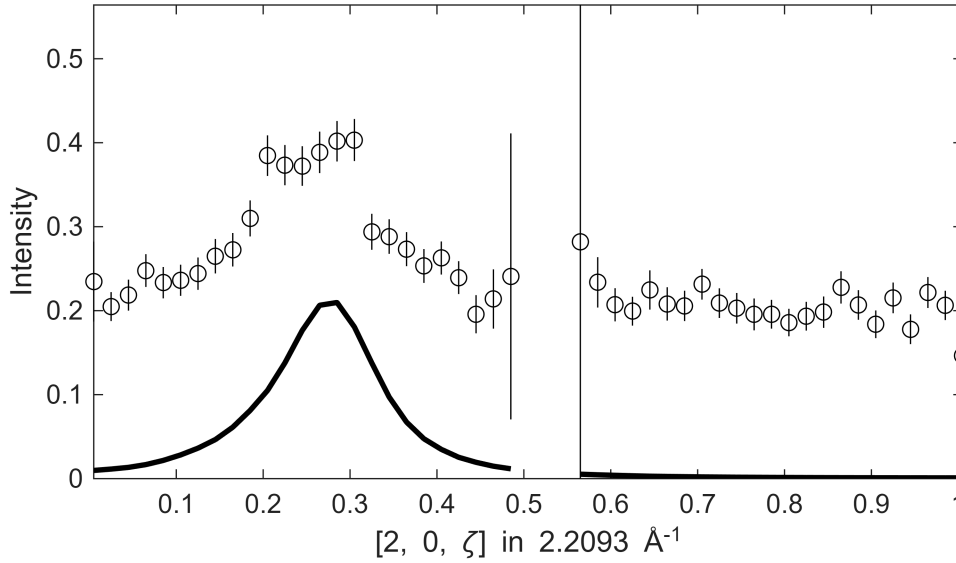


*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 99690 pixels -- processing data... -----> retained 65691 pixels

Number of points in input object: 99690
Fraction of object selected: 100.00% (= 99690 points) ; Access speed: 1.99GB/sec
Fraction of object retained: 65.90% (= 65691 points)

**** Total time in cut(sqw) : 0.0sec Including:
Data access time fraction: 19.2%; Transf. time frac.: 63.4%; Resulting pix preparation: 17.4%

Fe_ei401_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[2+\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, \eta, 0]$, $75 \leq E \leq 85$
 $\zeta = -0.005:0.02:1.015$ in $[2, 0, \zeta]$



**** En = 90±5

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested

*** Step 1 of 1; Read data for 102501 pixels -- processing data... -----> included 68373 pixels

Number of points in input object: 102501

Fraction of object selected: 100.00% (= 102501 points) ; Access speed: 1.88GB/sec

Fraction of object retained: 66.70% (= 68373 points)

**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 32.9%; Transf. time frac.: 61.1%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept

*** Step 1 of 1; Read data for 102501 pixels -- processing data... -----> retained 68373 pixels

Number of points in input object: 102501

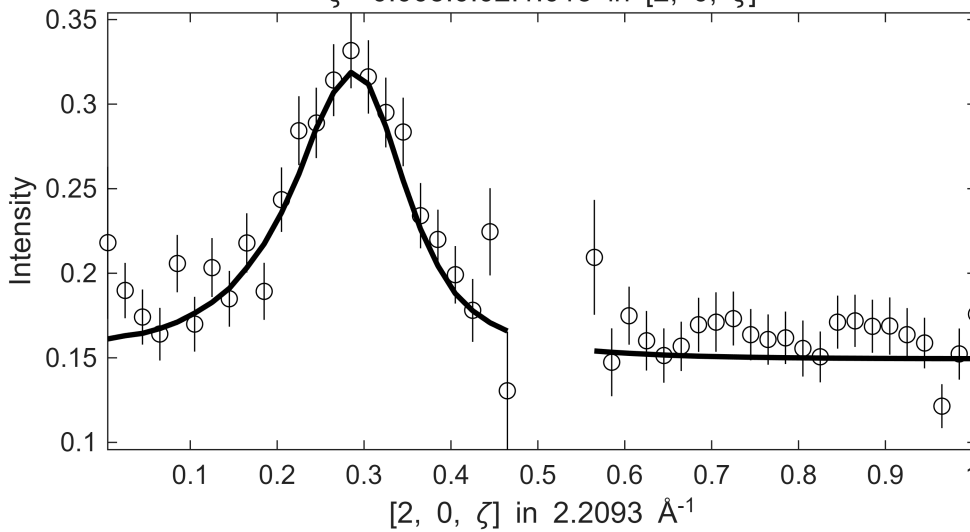
Fraction of object selected: 100.00% (= 102501 points) ; Access speed: 1.88GB/sec

Fraction of object retained: 66.70% (= 68373 points)

**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 21.5%; Transf. time frac.: 56.7%; Resulting pix preparation: 21.8%

Fe_ei401_align_sym3.sqw
 $S=0.83\pm0.099$; $J0=29\pm1.1$; $\gamma=40.6658\pm7.28193$
 $-0.1 \leq \xi \leq 0.1$ in $[2+\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, \eta, 0]$, $85 \leq E \leq 95$
 $\zeta=-0.005:0.02:1.015$ in $[2, 0, \zeta]$

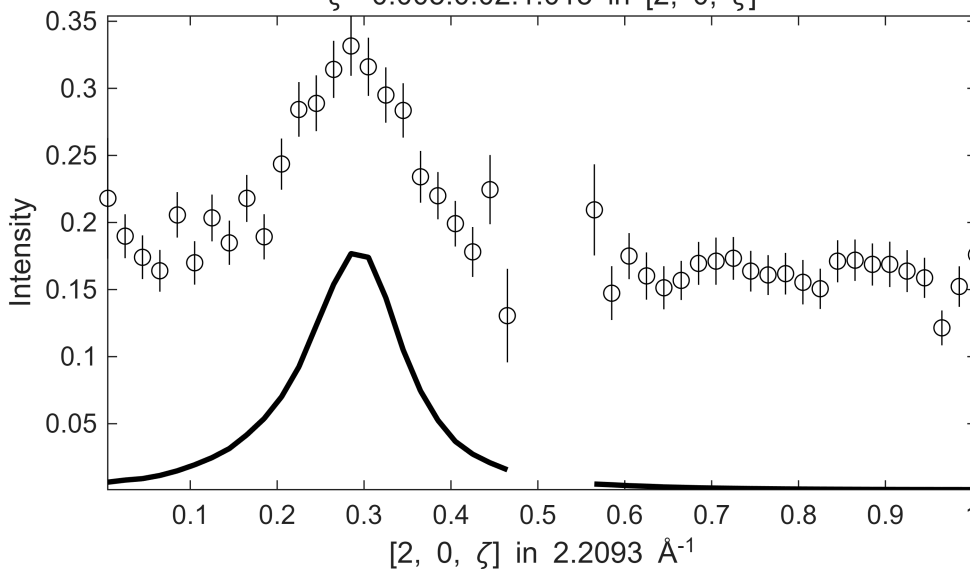


*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
 *** Step 1 of 1; Read data for 102501 pixels -- processing data... -----> retained 68373 pixels

Number of points in input object: 102501
 Fraction of object selected: 100.00% (= 102501 points) ; Access speed: 1.90GB/sec
 Fraction of object retained: 66.70% (= 68373 points)

**** Total time in cut(sqw) : 0.0sec Including:
 Data access time fraction: 20.7%; Transf. time frac.: 62.3%; Resulting pix preparation: 17.0%

Fe_ei401_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[2+\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, \eta, 0]$, $85 \leq E \leq 95$
 $\zeta=-0.005:0.02:1.015$ in $[2, 0, \zeta]$



 **** En = 100±5

 *** Cutting in memory sqw object; returning result in memory; No resulting pixels requested
 *** Step 1 of 1; Read data for 107412 pixels -- processing data... -----> included 71666 pixels

Number of points in input object: 107412
 Fraction of object selected: 100.00% (= 107412 points) ; Access speed: 1.92GB/sec
 Fraction of object retained: 66.72% (= 71666 points)

**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 33.8%; Transf. time frac.: 60.3%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 107412 pixels -- processing data... -----> retained 71666 pixels

Number of points in input object: 107412

Fraction of object selected: 100.00% (= 107412 points) ; Access speed: 1.65GB/sec

Fraction of object retained: 66.72% (= 71666 points)

**** Total time in cut(sqw) : 0.0sec Including:

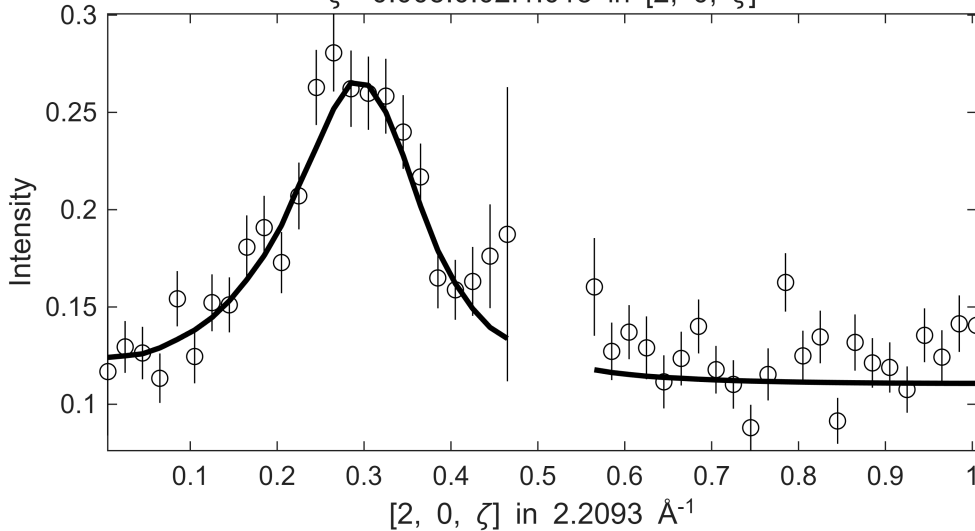
Data access time fraction: 22.0%; Transf. time frac.: 55.9%; Resulting pix preparation: 22.1%

Fe_ei401_align_sym3.sqw

S= 1 ± 0.14 ; J0= 33 ± 1.6 ; gamma= 57.9954 ± 11.8987

$-0.1 \leq \xi \leq 0.1$ in $[2+\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, \eta, 0]$, $95 \leq E \leq 105$

$\zeta = -0.005:0.02:1.015$ in $[2, 0, \zeta]$



*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept

*** Step 1 of 1; Read data for 107412 pixels -- processing data... -----> retained 71666 pixels

Number of points in input object: 107412

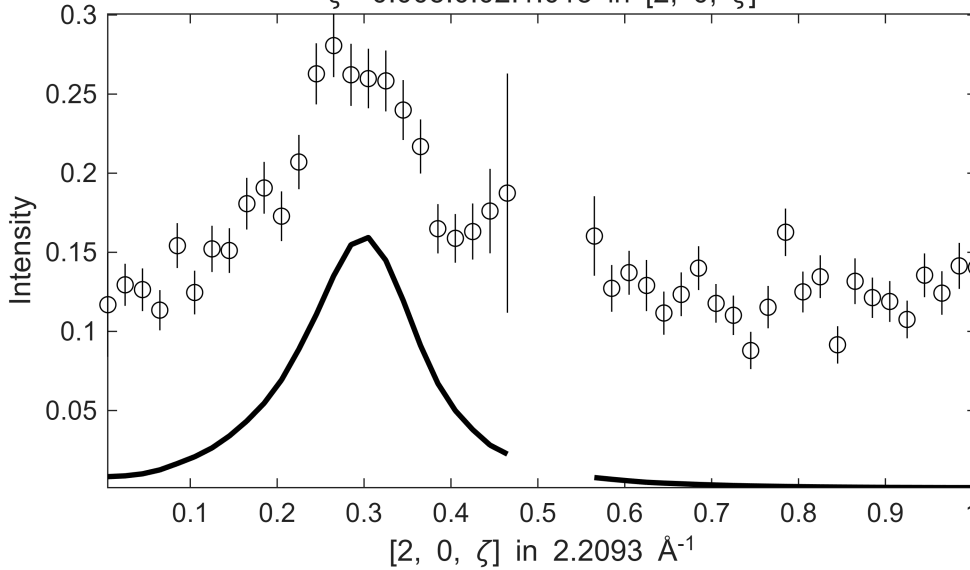
Fraction of object selected: 100.00% (= 107412 points) ; Access speed: 1.80GB/sec

Fraction of object retained: 66.72% (= 71666 points)

**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 22.4%; Transf. time frac.: 55.0%; Resulting pix preparation: 22.6%

Fe_ei401_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[2+\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, \eta, 0]$, $95 \leq E \leq 105$
 $\zeta = -0.005:0.02:1.015$ in $[2, 0, \zeta]$



**** En = 110±5

Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.4902

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested

*** Step 1 of 1; Read data for 114156 pixels -- processing data... -----> included 74770 pixels

Number of points in input object: 114156

Fraction of object selected: 100.00% (= 114156 points) ; Access speed: 1.95GB/sec

Fraction of object retained: 65.50% (= 74770 points)

**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 33.1%; Transf. time frac.: 60.6%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept

*** Step 1 of 1; Read data for 114144 pixels -- processing data... -----> retained 74758 pixels

Number of points in input object: 114144

Fraction of object selected: 100.00% (= 114144 points) ; Access speed: 1.67GB/sec

Fraction of object retained: 65.49% (= 74758 points)

**** Total time in cut(sqw) : 0.0sec Including:

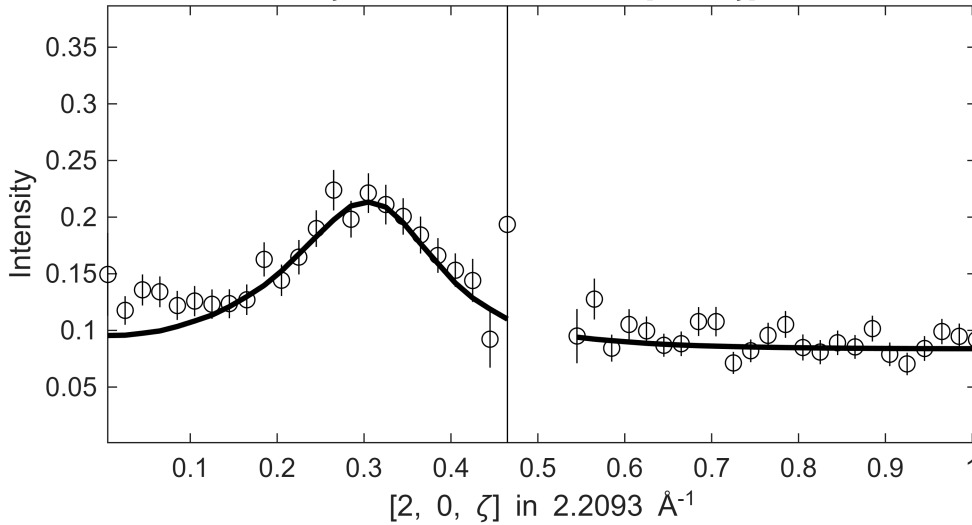
Data access time fraction: 20.5%; Transf. time frac.: 59.1%; Resulting pix preparation: 20.3%

Fe_ei401_align_sym3.sqw

S= 1.1± 0.2; J0= 36± 2.4; gamma=80.5074±20.2987

-0.1 ≤ ξ ≤ 0.1 in [2+ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, η, 0] , 105 ≤ E ≤ 115

ζ=-0.005:0.02:1.015 in [2, 0, ζ]



*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept

*** Step 1 of 1; Read data for 114156 pixels -- processing data... -----> retained 74770 pixels

Number of points in input object: 114156

Fraction of object selected: 100.00% (= 114156 points) ; Access speed: 1.63GB/sec

Fraction of object retained: 65.50% (= 74770 points)

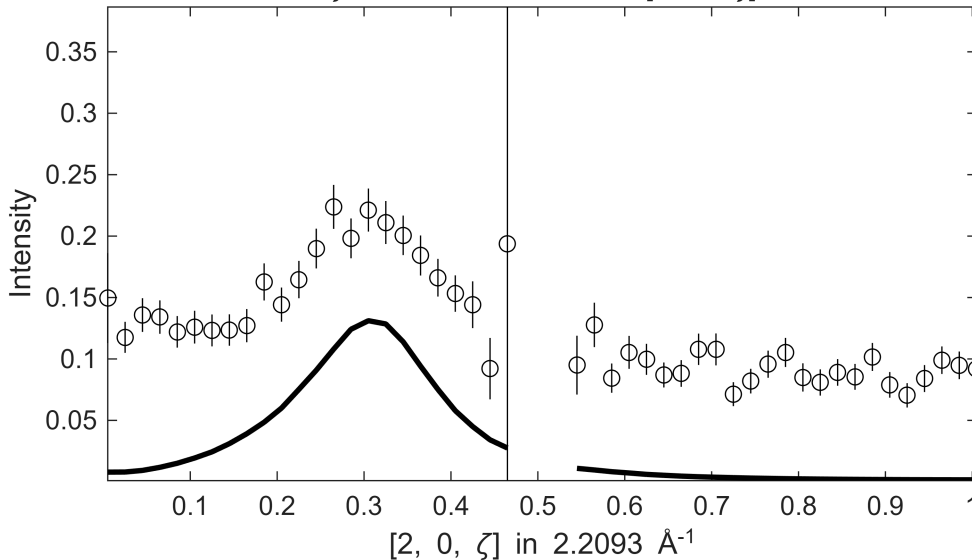
**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 20.3%; Transf. time frac.: 58.0%; Resulting pix preparation: 21.7%

Fe_ei401_align_sym3.sqw

-0.1 ≤ ξ ≤ 0.1 in [2+ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, η, 0] , 105 ≤ E ≤ 115

ζ=-0.005:0.02:1.015 in [2, 0, ζ]



**** En = 120±5

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested

*** Step 1 of 1; Read data for 124250 pixels -- processing data... -----> included 81071 pixels

Number of points in input object: 124250

Fraction of object selected: 100.00% (= 124250 points) ; Access speed: 1.89GB/sec

Fraction of object retained: 65.25% (= 81071 points)

**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 33.6%; Transf. time frac.: 60.9%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 124250 pixels -- processing data... -----> retained 81071 pixels

Number of points in input object: 124250

Fraction of object selected: 100.00% (= 124250 points) ; Access speed: 1.68GB/sec

Fraction of object retained: 65.25% (= 81071 points)

**** Total time in cut(sqw) : 0.0sec Including:

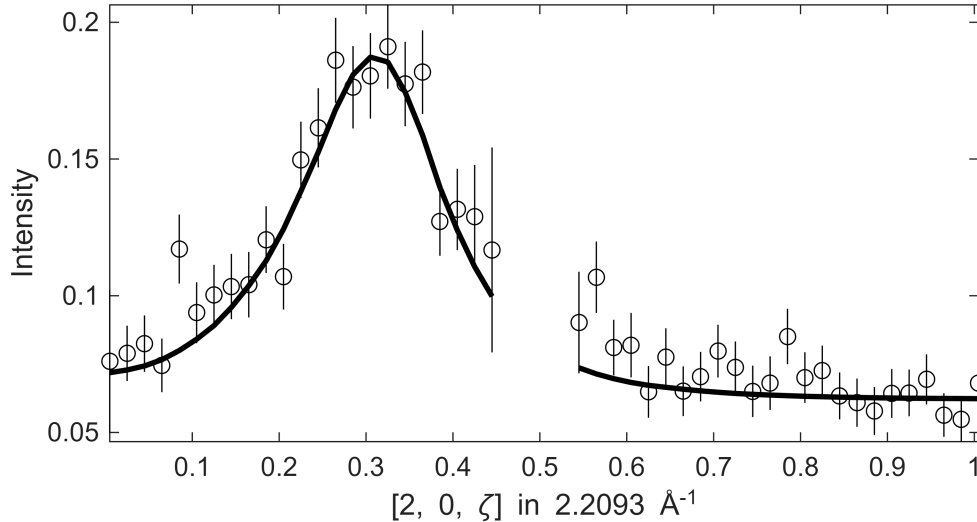
Data access time fraction: 22.1%; Transf. time frac.: 57.1%; Resulting pix preparation: 20.8%

Fe_ei401_align_sym3.sqw

S= 1.1±0.04; J0= 37±0.82; gamma=84.1933±2.14545

-0.1 ≤ ξ ≤ 0.1 in [2+ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, η, 0] , 115 ≤ E ≤ 125

ζ=-0.005:0.02:1.015 in [2, 0, ζ]



*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 124250 pixels -- processing data... -----> retained 81071 pixels

Number of points in input object: 124250

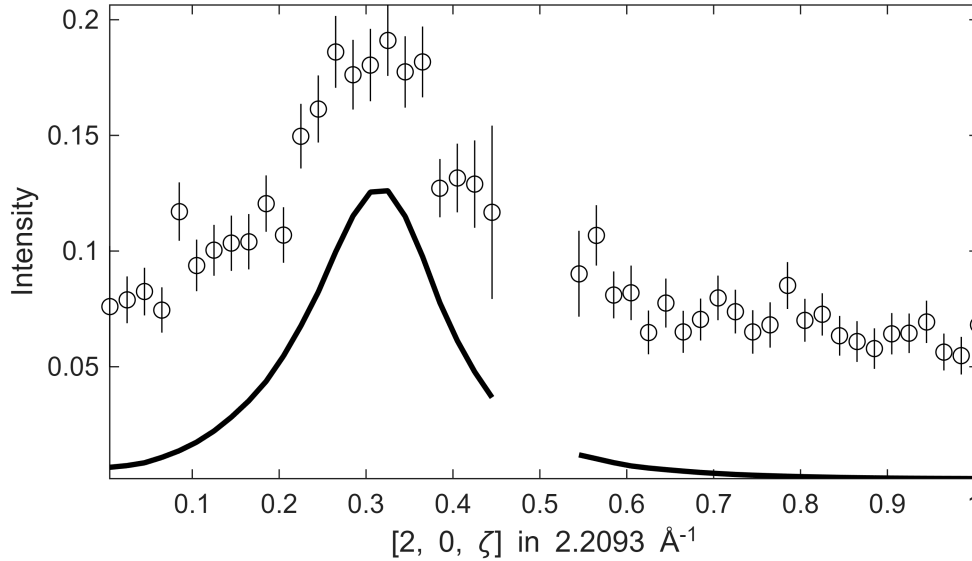
Fraction of object selected: 100.00% (= 124250 points) ; Access speed: 2.06GB/sec

Fraction of object retained: 65.25% (= 81071 points)

**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 19.4%; Transf. time frac.: 59.6%; Resulting pix preparation: 21.0%

Fe_ei401_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[2+\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, \eta, 0]$, $115 \leq E \leq 125$
 $\zeta = -0.005:0.02:1.015$ in $[2, 0, \zeta]$



**** En = 130±5

Warning: Dataset:2 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.98039

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested

*** Step 1 of 1; Read data for 141060 pixels -- processing data... -----> included 93097 pixels

Number of points in input object: 141060

Fraction of object selected: 100.00% (= 141060 points) ; Access speed: 1.96GB/sec

Fraction of object retained: 66.00% (= 93097 points)

**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 36.2%; Transf. time frac.: 59.3%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept

*** Step 1 of 1; Read data for 141019 pixels -- processing data... -----> retained 93056 pixels

Number of points in input object: 141019

Fraction of object selected: 100.00% (= 141019 points) ; Access speed: 1.55GB/sec

Fraction of object retained: 65.99% (= 93056 points)

**** Total time in cut(sqw) : 0.0sec Including:

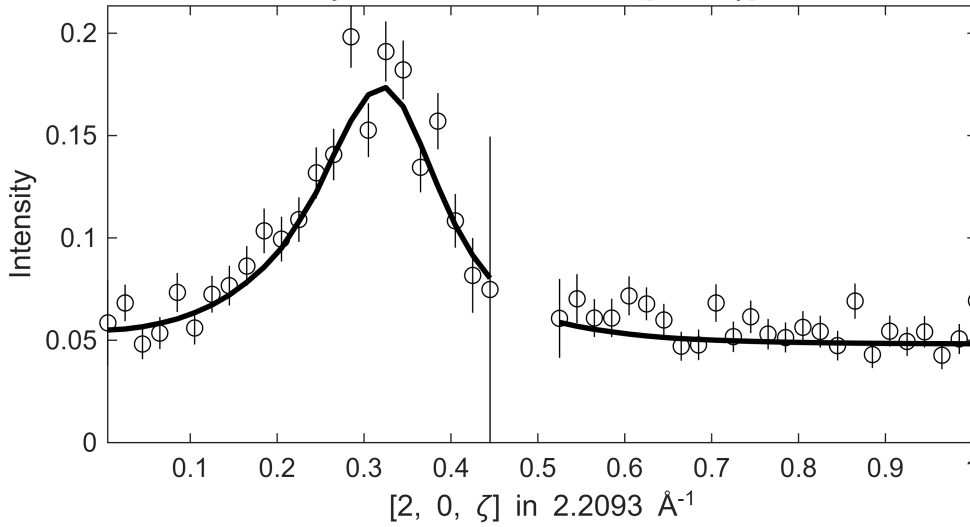
Data access time fraction: 26.4%; Transf. time frac.: 55.6%; Resulting pix preparation: 18.0%

Fe_ei401_align_sym3.sqw

S=0.91± 0.1; J0= 36± 1.1; gamma=63.2412±10.1154

-0.1 ≤ ξ ≤ 0.1 in [2+ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, η, 0] , 125 ≤ E ≤ 135

ζ=-0.005:0.02:1.015 in [2, 0, ζ]



*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
 *** Step 1 of 1; Read data for 141060 pixels -- processing data... -----> retained 93097 pixels

Number of points in input object: 141060

Fraction of object selected: 100.00% (= 141060 points) ; Access speed: 1.84GB/sec

Fraction of object retained: 66.00% (= 93097 points)

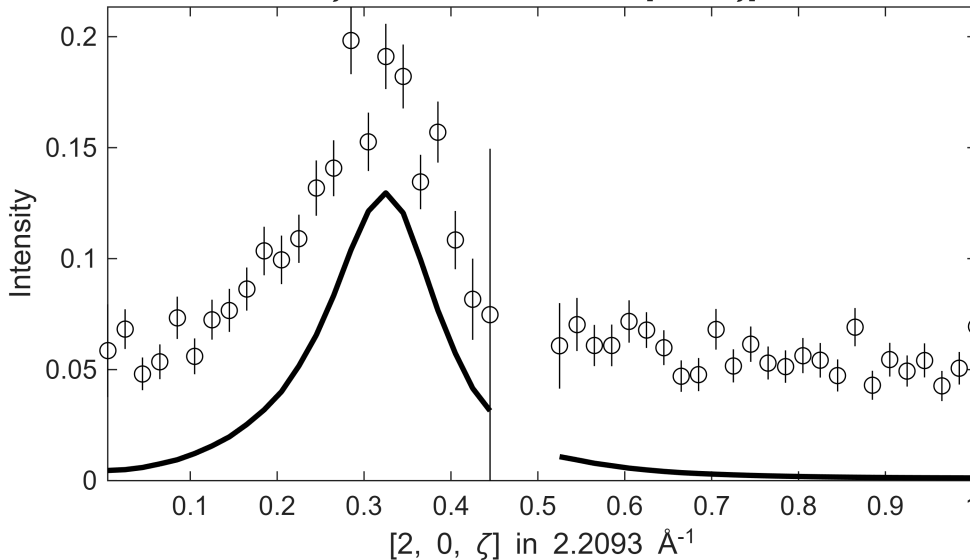
**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 23.9%; Transf. time frac.: 56.0%; Resulting pix preparation: 20.1%

Fe_ei401_align_sym3.sqw

-0.1 ≤ ξ ≤ 0.1 in [2+ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, η, 0] , 125 ≤ E ≤ 135

ζ=-0.005:0.02:1.015 in [2, 0, ζ]



**** En = 140±5

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested

*** Step 1 of 1; Read data for 152517 pixels -- processing data... -----> included 102029 pixels

Number of points in input object: 152517

Fraction of object selected: 100.00% (= 152517 points) ; Access speed: 1.68GB/sec

Fraction of object retained: 66.90% (= 102029 points)

**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 40.2%; Transf. time frac.: 55.5%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 152517 pixels -- processing data... -----> retained 102029 pixels

Number of points in input object: 152517

Fraction of object selected: 100.00% (= 152517 points) ; Access speed: 1.45GB/sec

Fraction of object retained: 66.90% (= 102029 points)

**** Total time in cut(sqw) : 0.0sec Including:

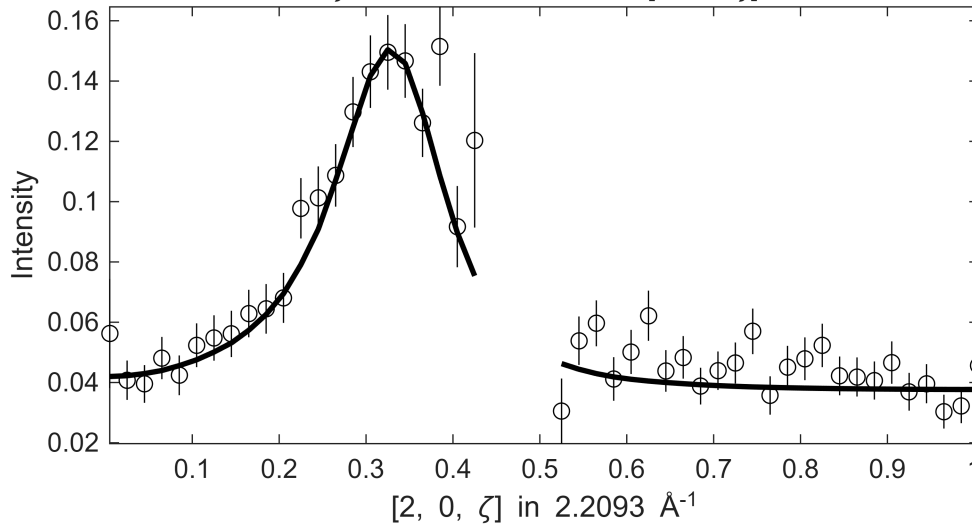
Data access time fraction: 26.9%; Transf. time frac.: 56.9%; Resulting pix preparation: 16.1%

Fe_ei401_align_sym3.sqw

$S=0.72\pm0.075$; $J_0=36\pm0.93$; $\gamma=53.1962\pm8.17817$

$-0.1 \leq \xi \leq 0.1$ in $[2+\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, \eta, 0]$, $135 \leq E \leq 145$

$\zeta=-0.005:0.02:1.015$ in $[2, 0, \zeta]$



*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 152517 pixels -- processing data... -----> retained 102029 pixels

Number of points in input object: 152517

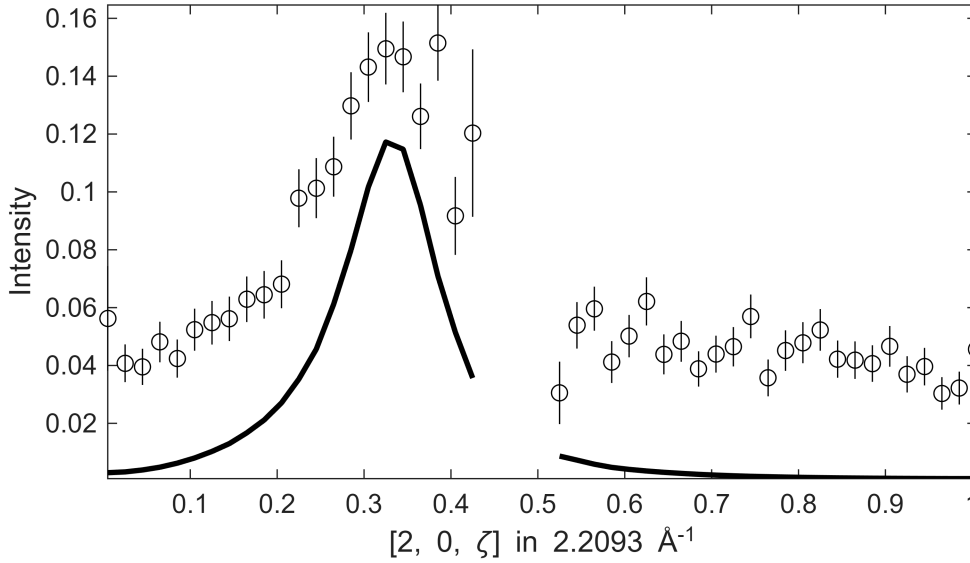
Fraction of object selected: 100.00% (= 152517 points) ; Access speed: 1.45GB/sec

Fraction of object retained: 66.90% (= 102029 points)

**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 17.9%; Transf. time frac.: 49.6%; Resulting pix preparation: 32.5%

Fe_ei401_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[2+\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, \eta, 0]$, $135 \leq E \leq 145$
 $\zeta = -0.005:0.02:1.015$ in $[2, 0, \zeta]$



**** En = 150±5

Warning: Dataset:2 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.98039

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested

*** Step 1 of 1; Read data for 157483 pixels -- processing data... -----> included 103824 pixels

Number of points in input object: 157483

Fraction of object selected: 100.00% (= 157483 points) ; Access speed: 1.66GB/sec

Fraction of object retained: 65.93% (= 103824 points)

**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 36.5%; Transf. time frac.: 59.9%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept

*** Step 1 of 1; Read data for 157447 pixels -- processing data... -----> retained 103793 pixels

Number of points in input object: 157447

Fraction of object selected: 100.00% (= 157447 points) ; Access speed: 1.50GB/sec

Fraction of object retained: 65.92% (= 103793 points)

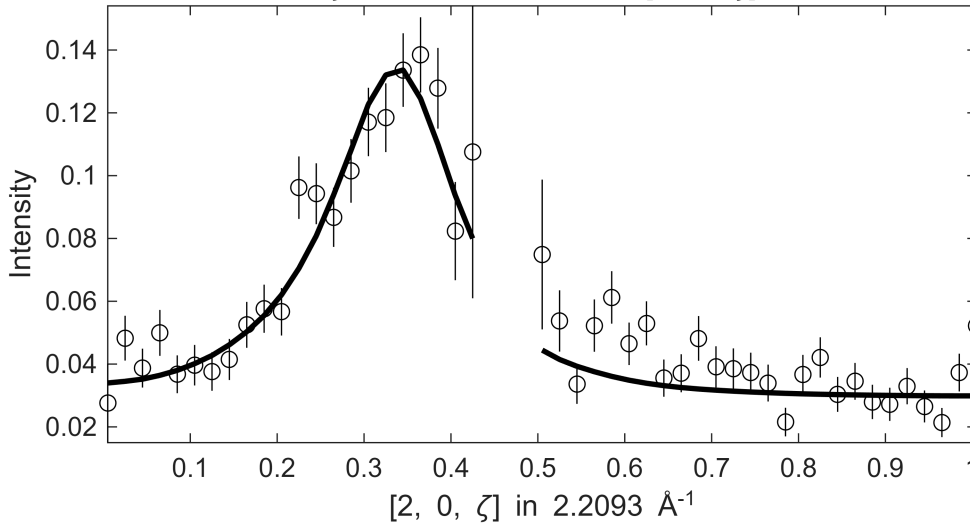
**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 27.9%; Transf. time frac.: 48.7%; Resulting pix preparation: 23.4%

Fe_ei401_align_sym3.sqw

S=0.84±0.034; J0= 38±0.91; gamma=69.7381±2.1259

-0.1 ≤ ξ ≤ 0.1 in [2+ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, η, 0] , 145 ≤ E ≤ 155
ζ=-0.005:0.02:1.015 in [2, 0, ζ]

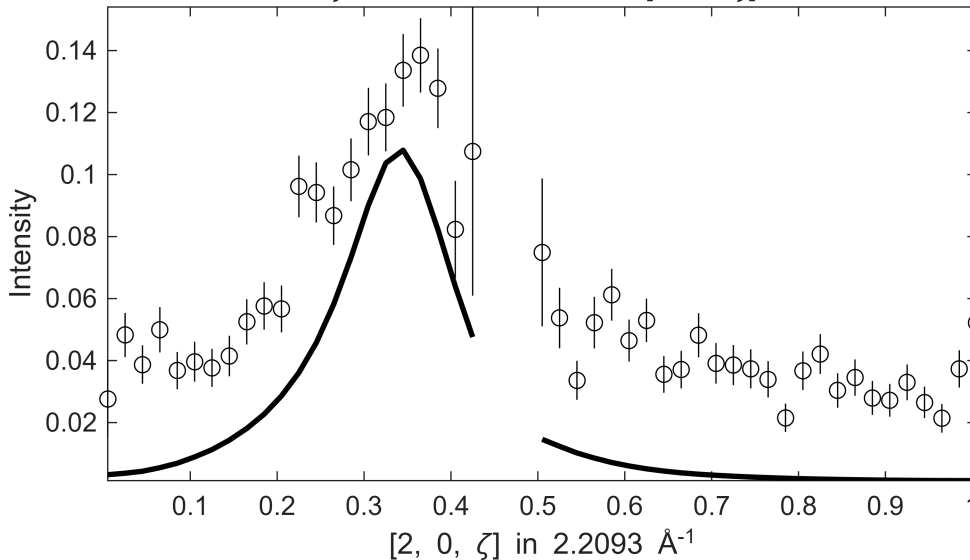


```
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 157483 pixels -- processing data... -----> retained 103824 pixels
-----
Number of points in input object: 157483
  Fraction of object selected: 100.00% (= 157483 points) ; Access speed: 1.52GB/sec
  Fraction of object retained: 65.93% (= 103824 points)

**** Total time in cut(sqw) : 0.0sec Including:
    Data access time fraction: 28.5%; Transf. time frac.: 50.1%; Resulting pix preparation: 21.4%
-----
```

Fe_ei401_align_sym3.sqw

-0.1 ≤ ξ ≤ 0.1 in [2+ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, η, 0] , 145 ≤ E ≤ 155
ζ=-0.005:0.02:1.015 in [2, 0, ζ]



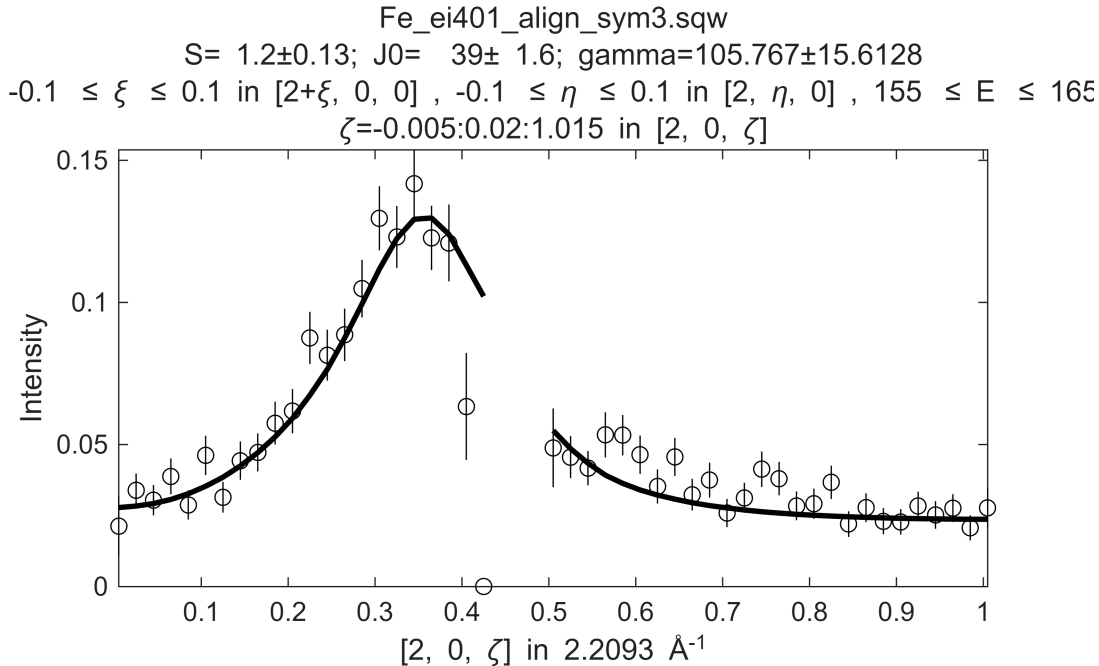
```
*****
**** En = 160±5
*****
Warning: Dataset:2 points with zero, negative or undefined (infinite or NaN) error bars have been removed
from fit, which is 0.98039
*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested
*** Step 1 of 1; Read data for 169690 pixels -- processing data... -----> included 111655 pixels
-----
Number of points in input object: 169690
  Fraction of object selected: 100.00% (= 169690 points) ; Access speed: 1.81GB/sec
  Fraction of object retained: 65.80% (= 111655 points)
```

```

**** Total time in cut(sqw)   :      0.0sec Including:
      Data access time fraction: 34.6%; Transf. time frac.: 60.9%; Resulting pix preparation:  0.0%
-----
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 169499 pixels -- processing data... -----> retained 111464 pixels
-----
Number of points in input object: 169499
      Fraction of object selected: 100.00% (= 169499 points) ; Access speed: 1.84GB/sec
      Fraction of object retained: 65.76% (= 111464 points)

**** Total time in cut(sqw)   :      0.0sec Including:
      Data access time fraction: 25.3%; Transf. time frac.: 55.0%; Resulting pix preparation: 19.6%
-----

```



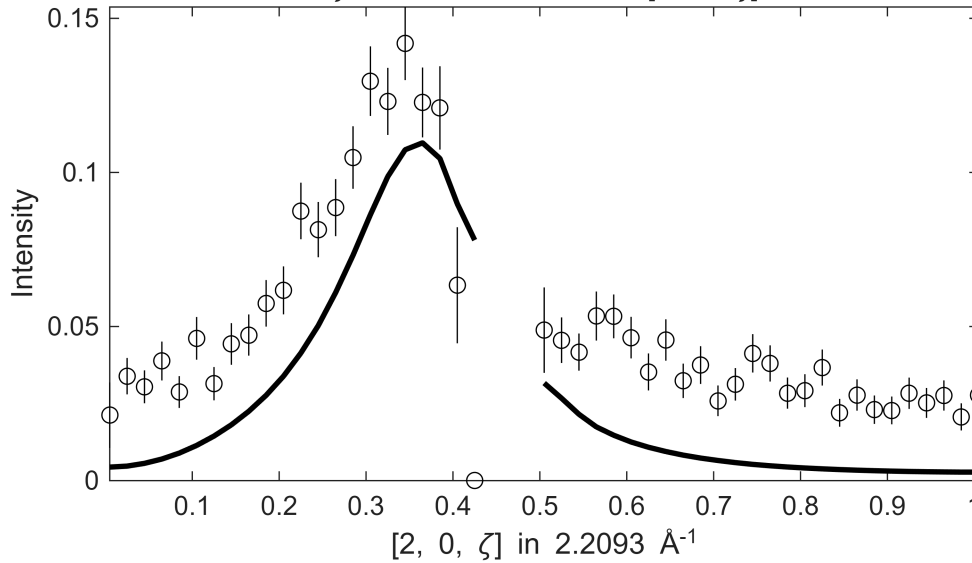
```

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 169690 pixels -- processing data... -----> retained 111655 pixels
-----
Number of points in input object: 169690
      Fraction of object selected: 100.00% (= 169690 points) ; Access speed: 1.55GB/sec
      Fraction of object retained: 65.80% (= 111655 points)

**** Total time in cut(sqw)   :      0.0sec Including:
      Data access time fraction: 24.7%; Transf. time frac.: 55.5%; Resulting pix preparation: 19.8%
-----

```

Fe_ei401_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[2+\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, \eta, 0]$, $155 \leq E \leq 165$
 $\zeta = -0.005:0.02:1.015$ in $[2, 0, \zeta]$



**** En = 170±5

Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.4902

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested

*** Step 1 of 1; Read data for 191730 pixels -- processing data... -----> included 126132 pixels

Number of points in input object: 191730

Fraction of object selected: 100.00% (= 191730 points) ; Access speed: 1.83GB/sec

Fraction of object retained: 65.79% (= 126132 points)

**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 40.0%; Transf. time frac.: 56.8%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept

*** Step 1 of 1; Read data for 191536 pixels -- processing data... -----> retained 126031 pixels

Number of points in input object: 191536

Fraction of object selected: 100.00% (= 191536 points) ; Access speed: 1.77GB/sec

Fraction of object retained: 65.80% (= 126031 points)

**** Total time in cut(sqw) : 0.0sec Including:

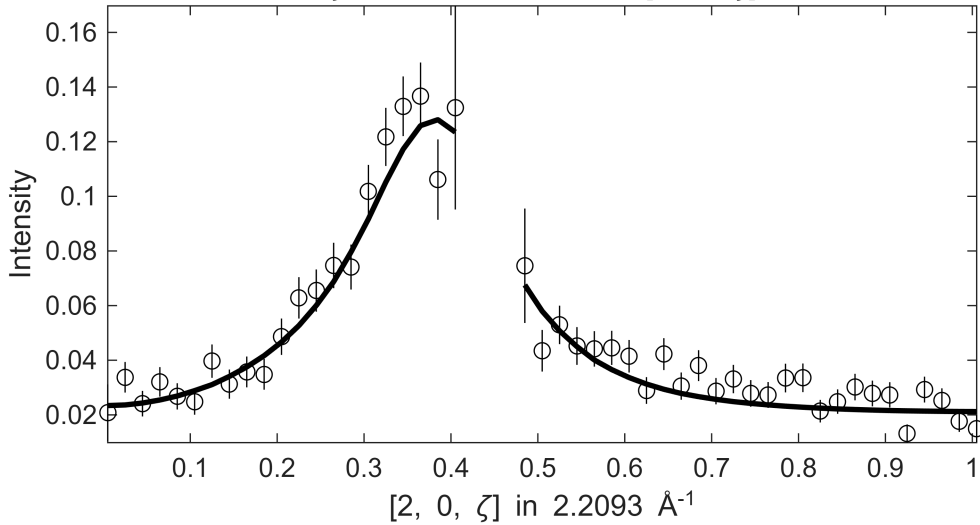
Data access time fraction: 26.4%; Transf. time frac.: 54.9%; Resulting pix preparation: 18.7%

Fe_ei401_align_sym3.sqw

S= 1.1±0.099; J0= 36± 1.3; gamma=89.546±12.9768

-0.1 ≤ ξ ≤ 0.1 in [2+ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, η, 0] , 165 ≤ E ≤ 175

ζ=-0.005:0.02:1.015 in [2, 0, ζ]



*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
 *** Step 1 of 1; Read data for 191730 pixels -- processing data... -----> retained 126132 pixels

Number of points in input object: 191730

Fraction of object selected: 100.00% (= 191730 points) ; Access speed: 1.89GB/sec

Fraction of object retained: 65.79% (= 126132 points)

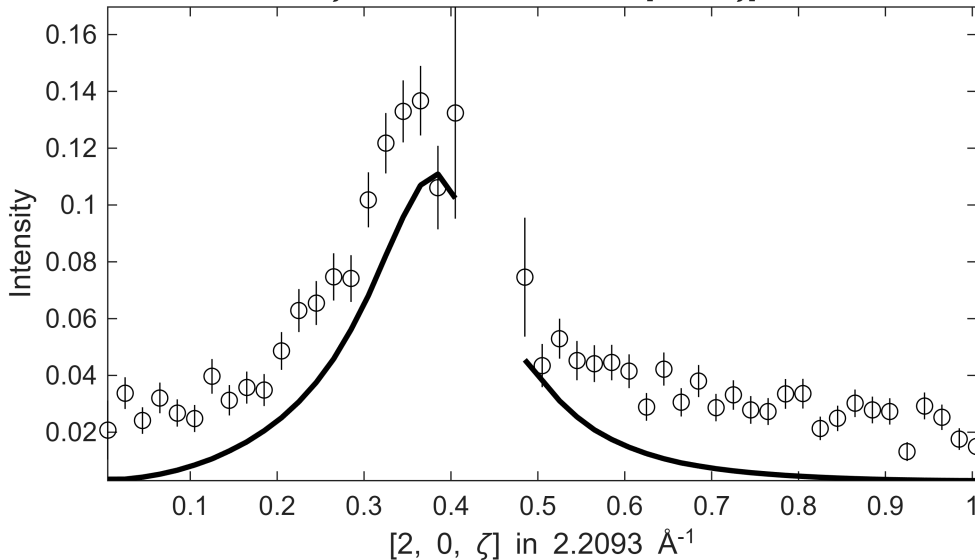
**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 26.6%; Transf. time frac.: 53.2%; Resulting pix preparation: 20.2%

Fe_ei401_align_sym3.sqw

-0.1 ≤ ξ ≤ 0.1 in [2+ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, η, 0] , 165 ≤ E ≤ 175

ζ=-0.005:0.02:1.015 in [2, 0, ζ]



**** En = 180±5

Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.4902

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested

*** Step 1 of 1; Read data for 234555 pixels -- processing data... -----> included 149526 pixels

Number of points in input object: 234555

Fraction of object selected: 100.00% (= 234555 points) ; Access speed: 1.64GB/sec

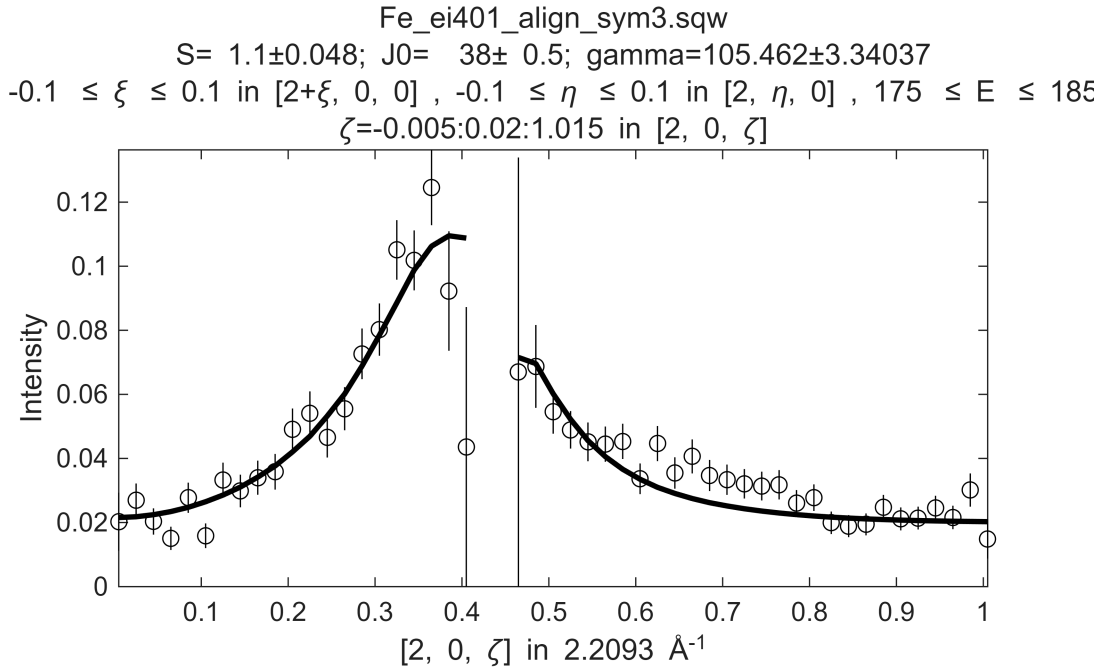
Fraction of object retained: 63.75% (= 149526 points)

```

**** Total time in cut(sqw)   :      0.0sec Including:
      Data access time fraction: 38.9%; Transf. time frac.: 58.0%; Resulting pix preparation:  0.0%
-----
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 234551 pixels -- processing data... -----> retained 149522 pixels
-----
Number of points in input object: 234551
      Fraction of object selected: 100.00% (= 234551 points) ; Access speed: 1.68GB/sec
      Fraction of object retained: 63.75% (= 149522 points)

**** Total time in cut(sqw)   :      0.0sec Including:
      Data access time fraction: 28.8%; Transf. time frac.: 54.5%; Resulting pix preparation: 16.7%
-----

```



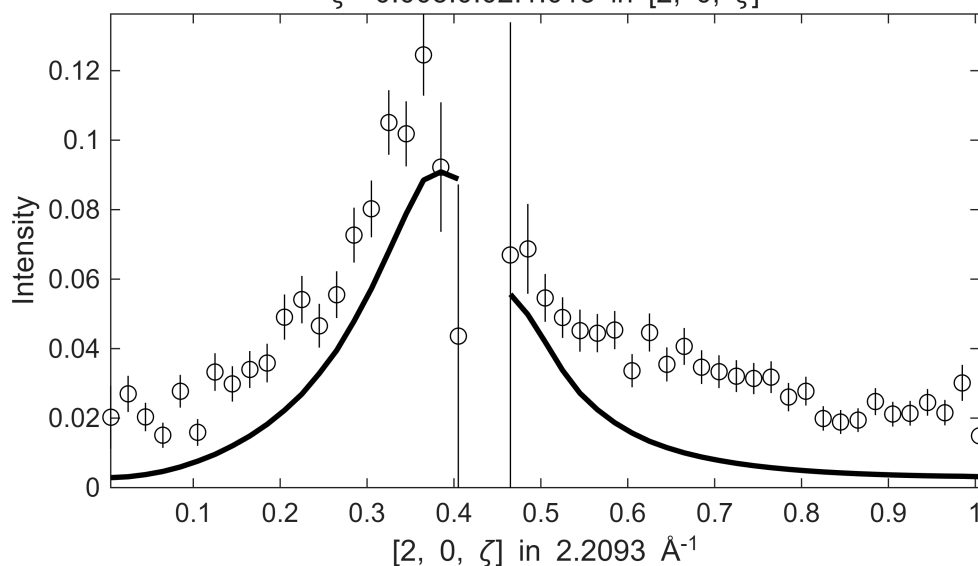
```

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 234555 pixels -- processing data... -----> retained 149526 pixels
-----
Number of points in input object: 234555
      Fraction of object selected: 100.00% (= 234555 points) ; Access speed: 1.58GB/sec
      Fraction of object retained: 63.75% (= 149526 points)

**** Total time in cut(sqw)   :      0.0sec Including:
      Data access time fraction: 29.1%; Transf. time frac.: 54.7%; Resulting pix preparation: 16.1%
-----

```

Fe_ei401_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[2+\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, \eta, 0]$, $175 \leq E \leq 185$
 $\zeta = -0.005:0.02:1.015$ in $[2, 0, \zeta]$



**** En = 190±5

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested

*** Step 1 of 1; Read data for 315109 pixels -- processing data... -----> included 197177 pixels

Number of points in input object: 315109

Fraction of object selected: 100.00% (= 315109 points) ; Access speed: 1.68GB/sec

Fraction of object retained: 62.57% (= 197177 points)

**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 46.3%; Transf. time frac.: 51.1%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept

*** Step 1 of 1; Read data for 315109 pixels -- processing data... -----> retained 197177 pixels

Number of points in input object: 315109

Fraction of object selected: 100.00% (= 315109 points) ; Access speed: 1.93GB/sec

Fraction of object retained: 62.57% (= 197177 points)

**** Total time in cut(sqw) : 0.0sec Including:

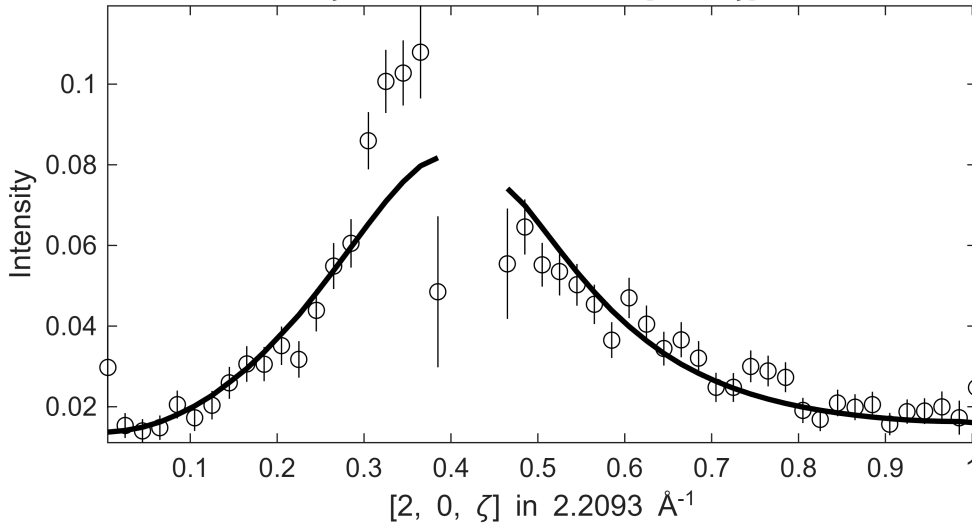
Data access time fraction: 29.5%; Transf. time frac.: 54.1%; Resulting pix preparation: 16.4%

Fe_ei401_align_sym3.sqw

S= 2.5±0.88; J0= 55± 14; gamma=402.145±231.381

-0.1 ≤ ξ ≤ 0.1 in [2+ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, η, 0] , 185 ≤ E ≤ 195

ζ=-0.005:0.02:1.015 in [2, 0, ζ]



*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
 *** Step 1 of 1; Read data for 315109 pixels -- processing data... -----> retained 197177 pixels

Number of points in input object: 315109

Fraction of object selected: 100.00% (= 315109 points) ; Access speed: 1.66GB/sec

Fraction of object retained: 62.57% (= 197177 points)

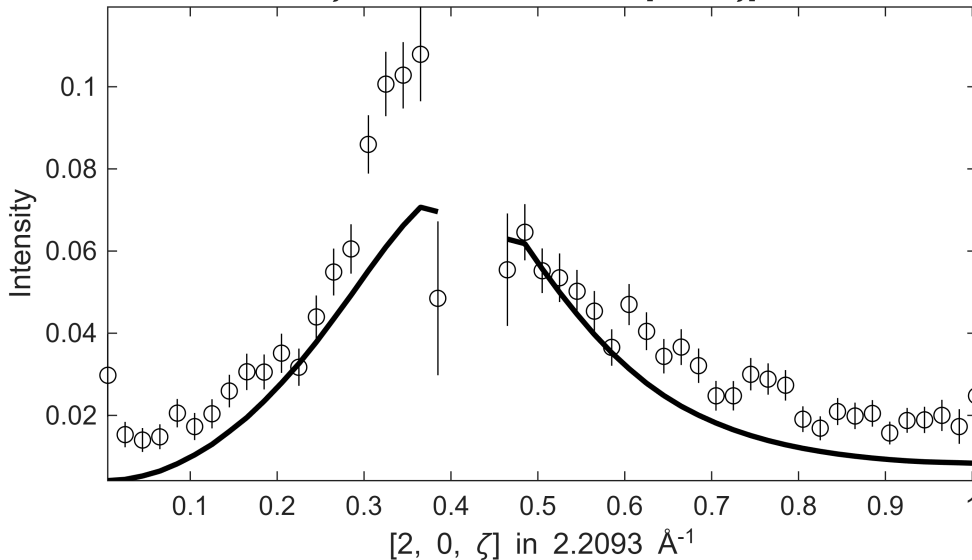
**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 28.8%; Transf. time frac.: 54.0%; Resulting pix preparation: 17.1%

Fe_ei401_align_sym3.sqw

-0.1 ≤ ξ ≤ 0.1 in [2+ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, η, 0] , 185 ≤ E ≤ 195

ζ=-0.005:0.02:1.015 in [2, 0, ζ]



**** En = 200±5

Warning: Dataset:2 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.98039

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested

*** Step 1 of 1; Read data for 466724 pixels -- processing data... -----> included 311912 pixels

Number of points in input object: 466724

Fraction of object selected: 100.00% (= 466724 points) ; Access speed: 1.60GB/sec

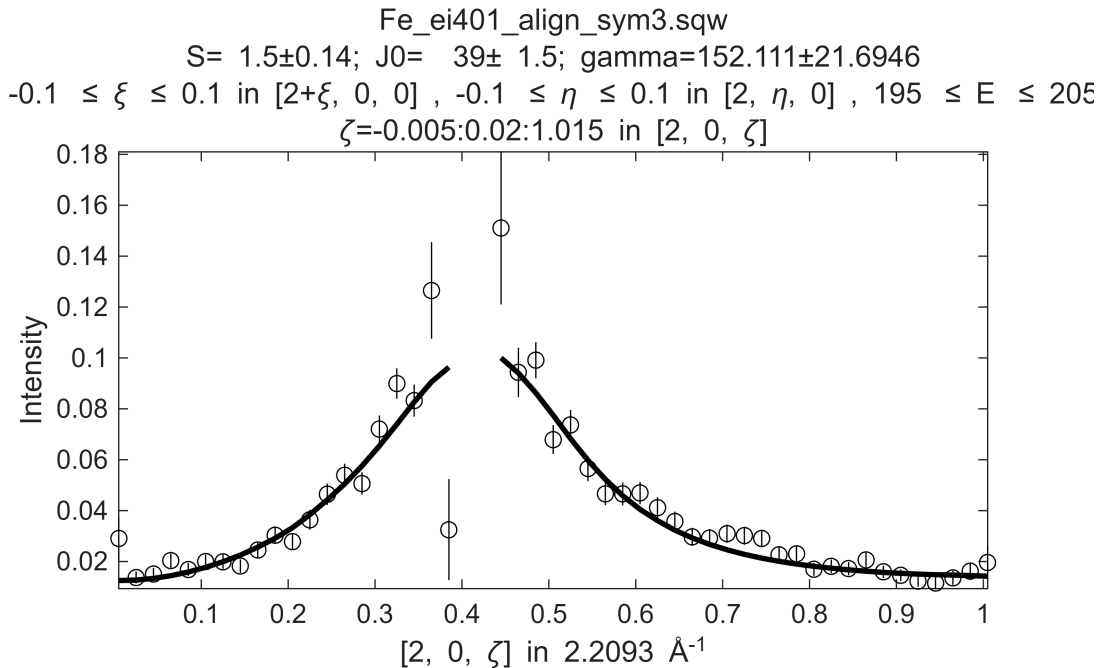
Fraction of object retained: 66.83% (= 311912 points)

```

**** Total time in cut(sqw)   :      0.0sec Including:
      Data access time fraction:  46.4%; Transf. time frac.:  51.9%; Resulting pix preparation:  0.0%
-----
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for  466575 pixels -- processing data... -----> retained    311763 pixels
-----
Number of points in input object: 466575
      Fraction of object selected:  100.00% (=   466575 points) ; Access speed:      1.87GB/sec
      Fraction of object retained:   66.82% (=   311763 points)

**** Total time in cut(sqw)   :      0.1sec Including:
      Data access time fraction:  32.7%; Transf. time frac.:  51.5%; Resulting pix preparation: 15.8%
-----

```



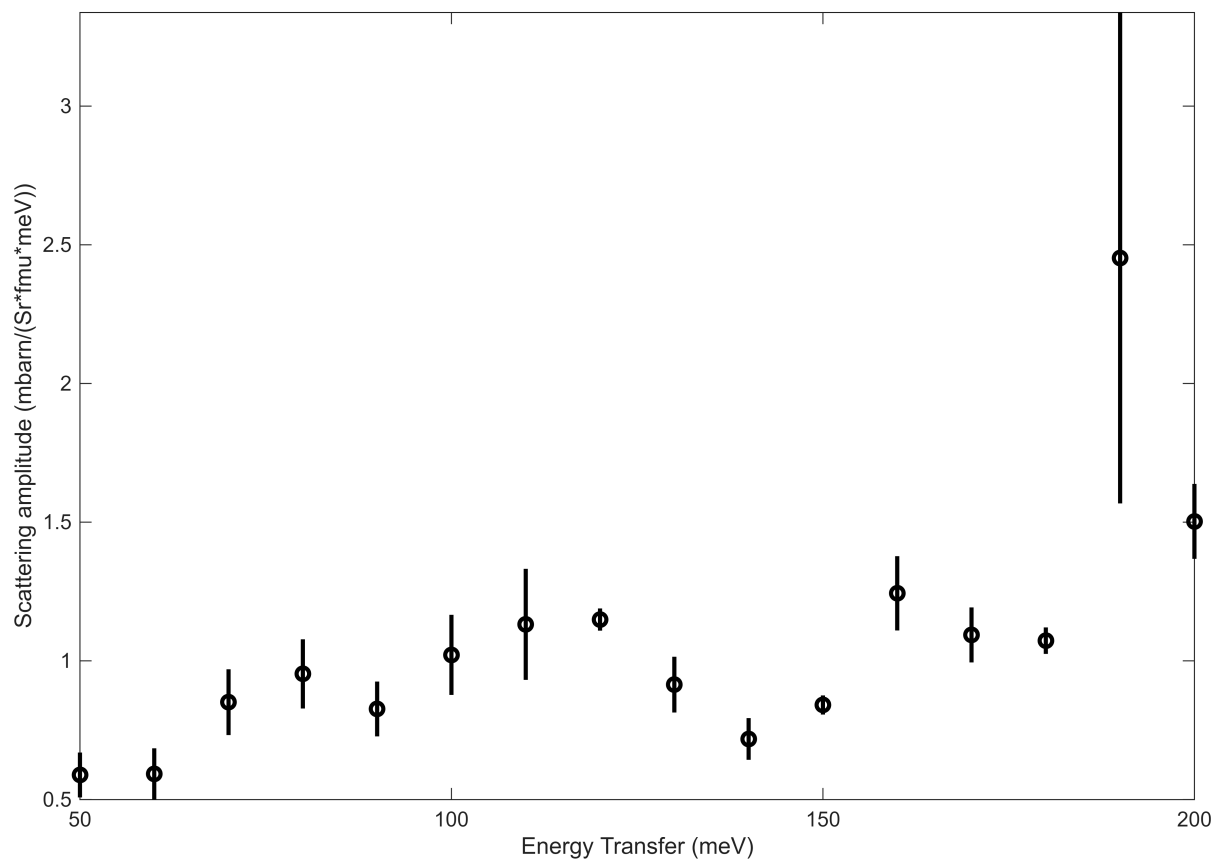
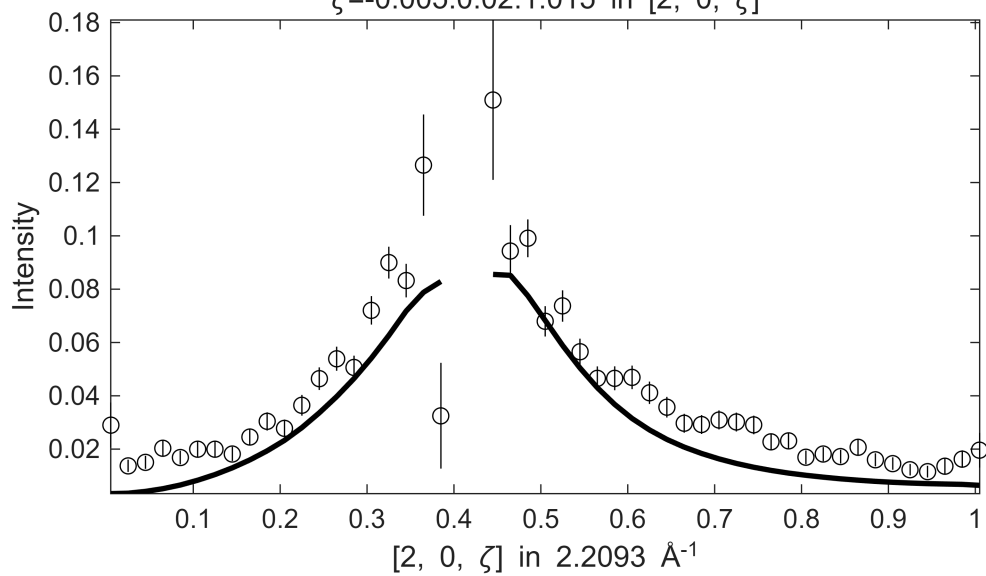
```

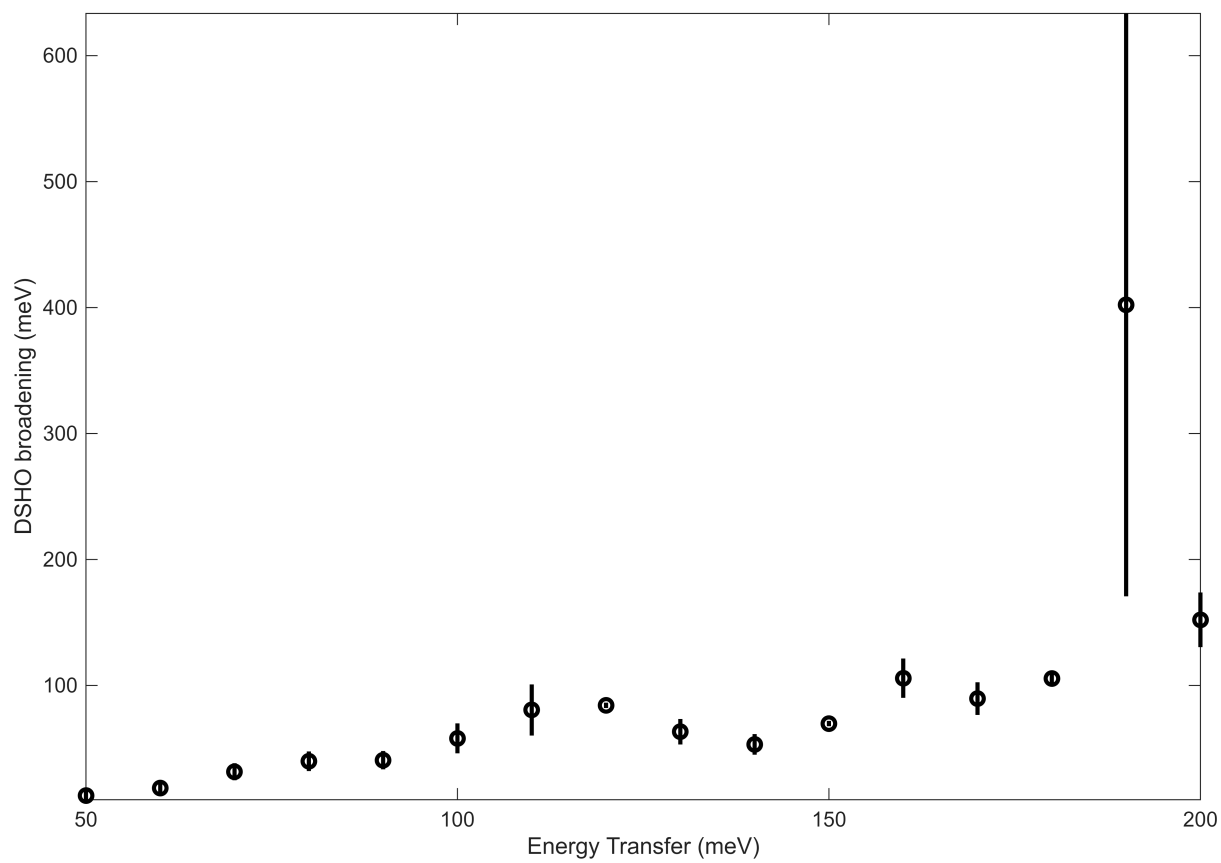
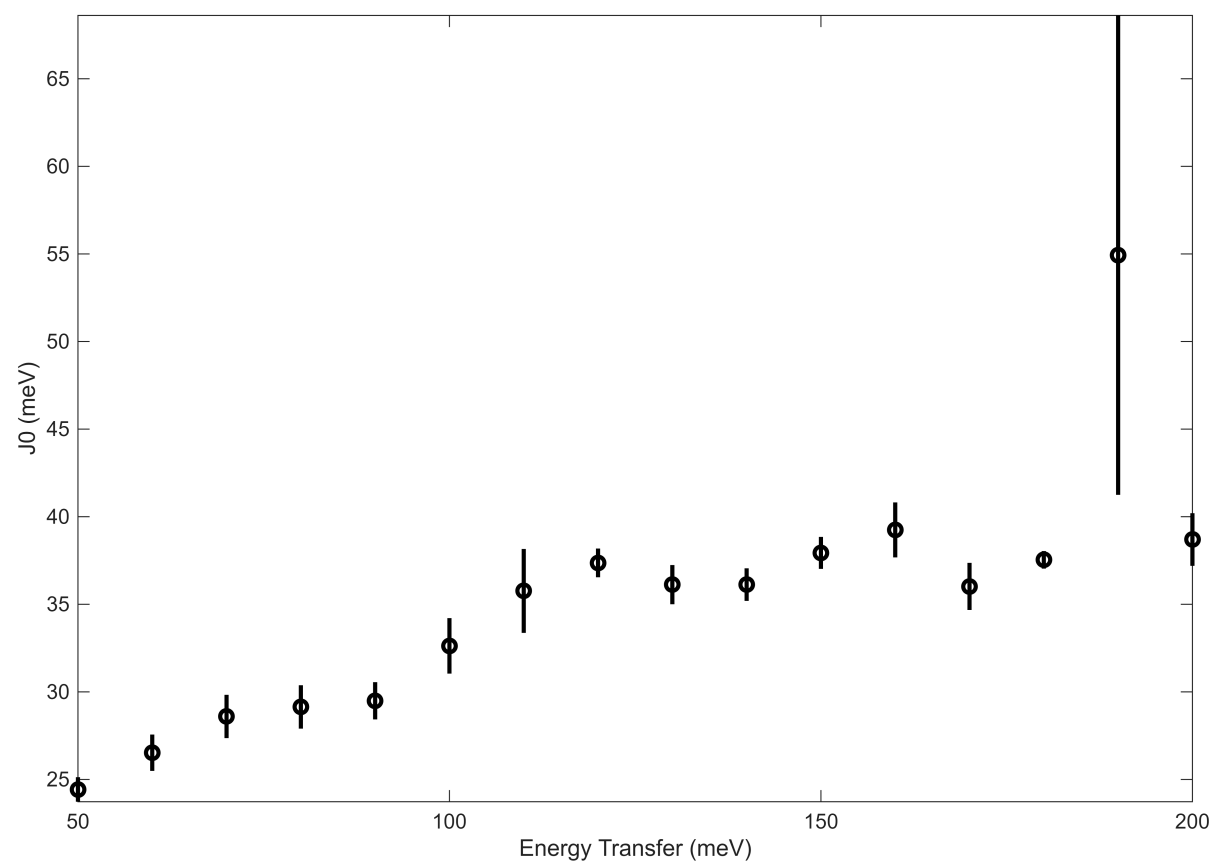
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for  466724 pixels -- processing data... -----> retained    311912 pixels
-----
Number of points in input object: 466724
      Fraction of object selected:  100.00% (=   466724 points) ; Access speed:      1.96GB/sec
      Fraction of object retained:   66.83% (=   311912 points)

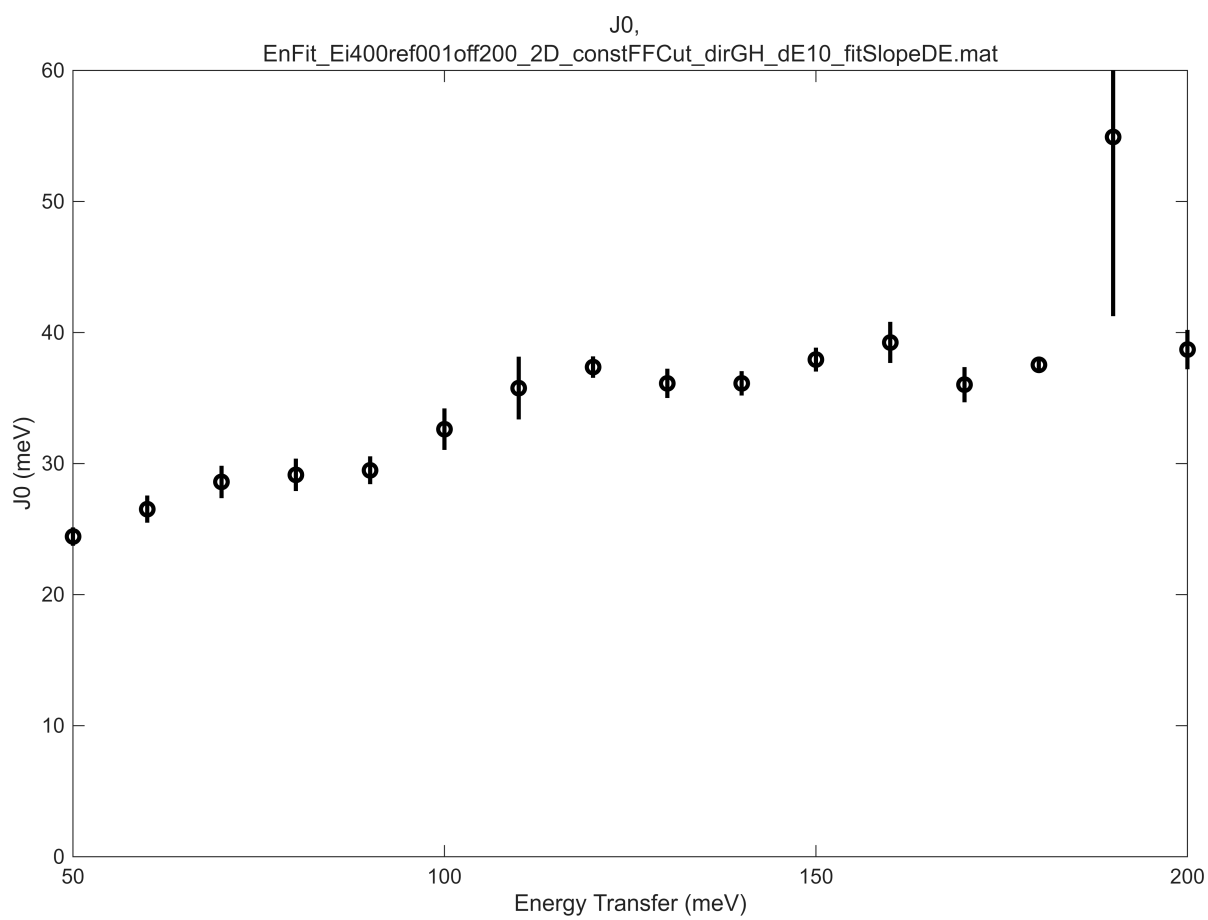
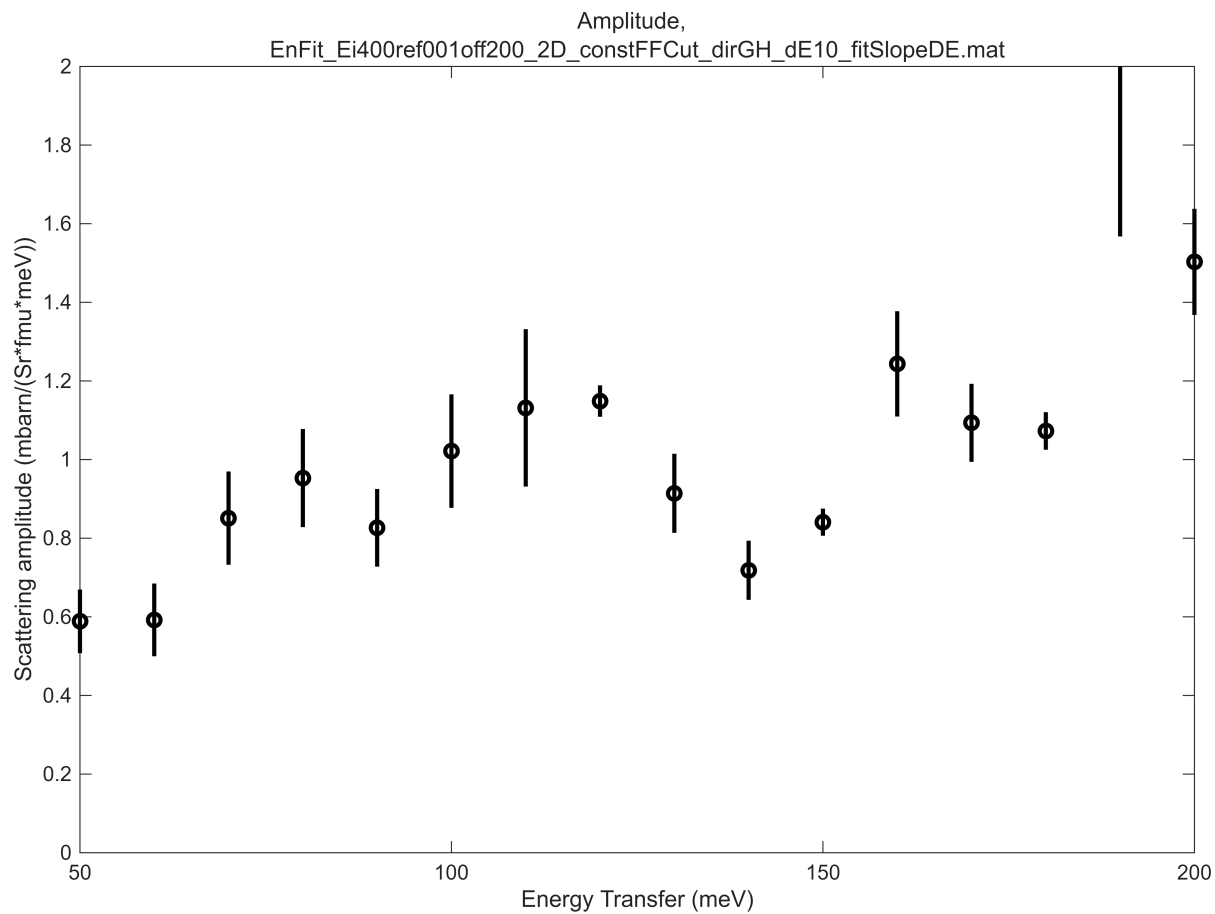
**** Total time in cut(sqw)   :      0.1sec Including:
      Data access time fraction:  31.2%; Transf. time frac.:  53.0%; Resulting pix preparation: 15.8%
-----

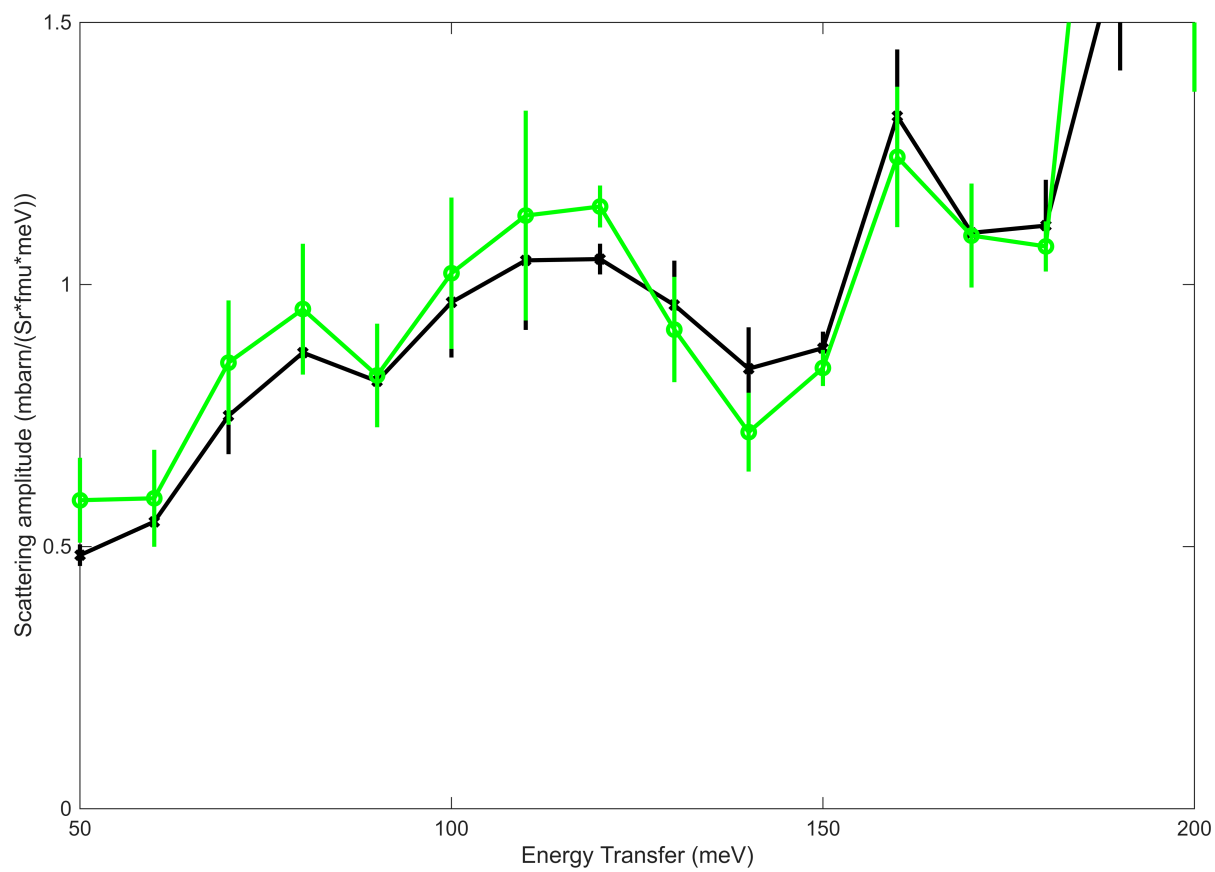
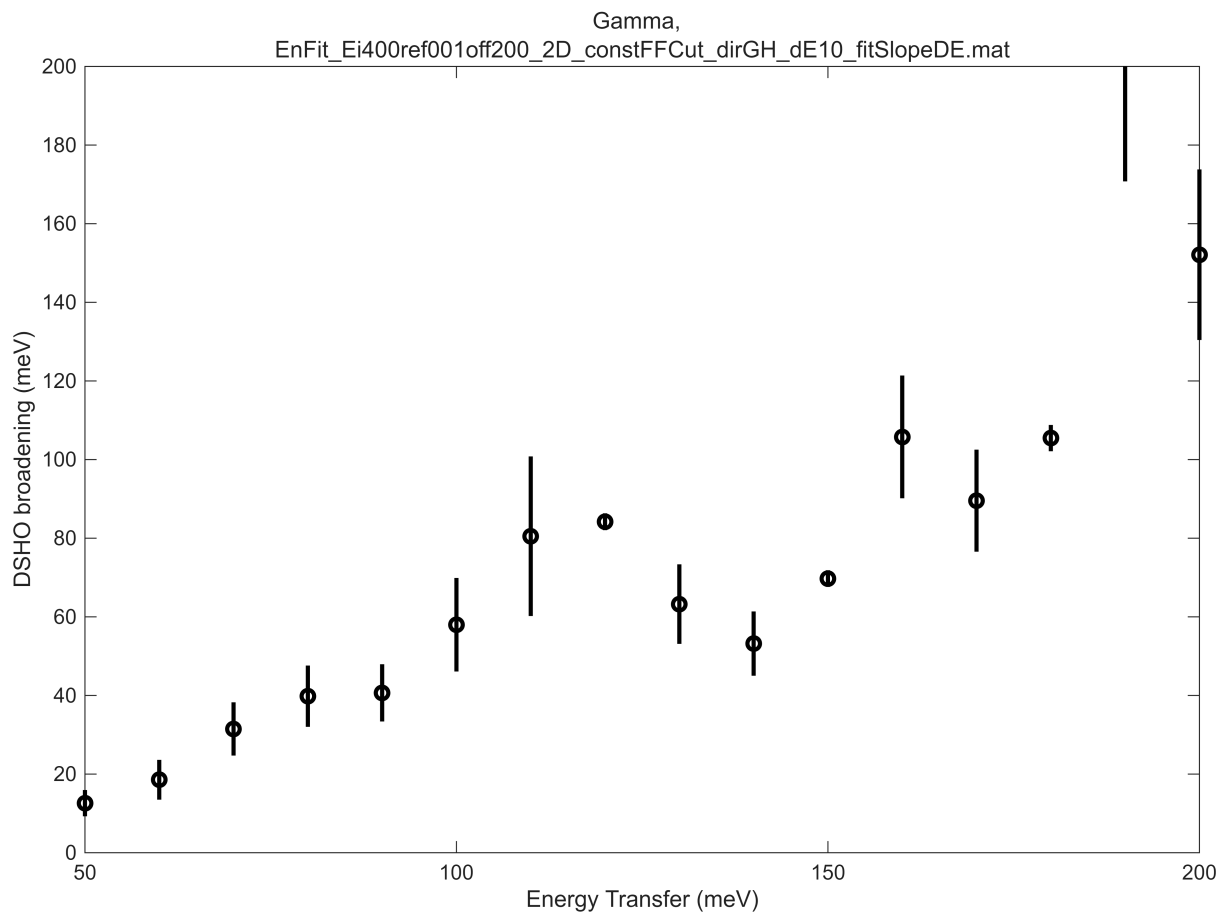
```

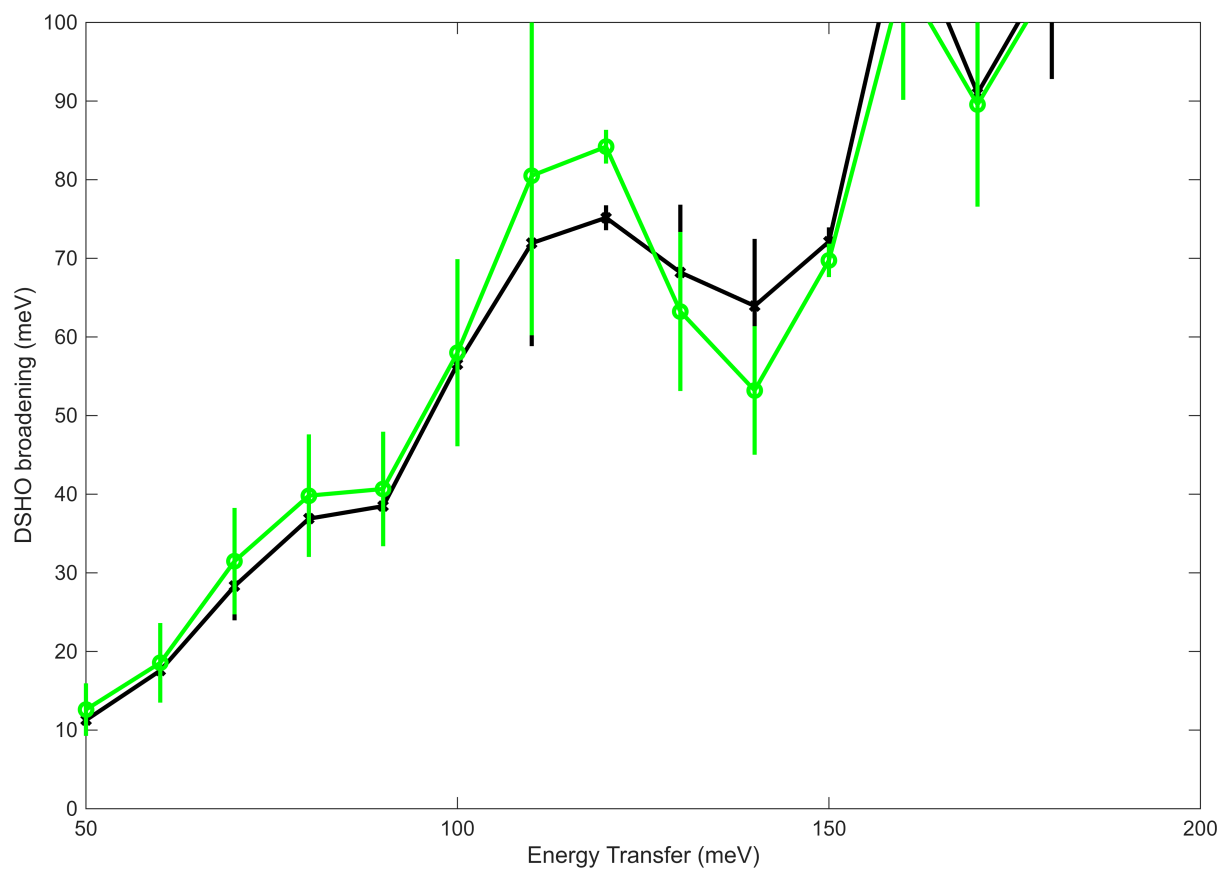
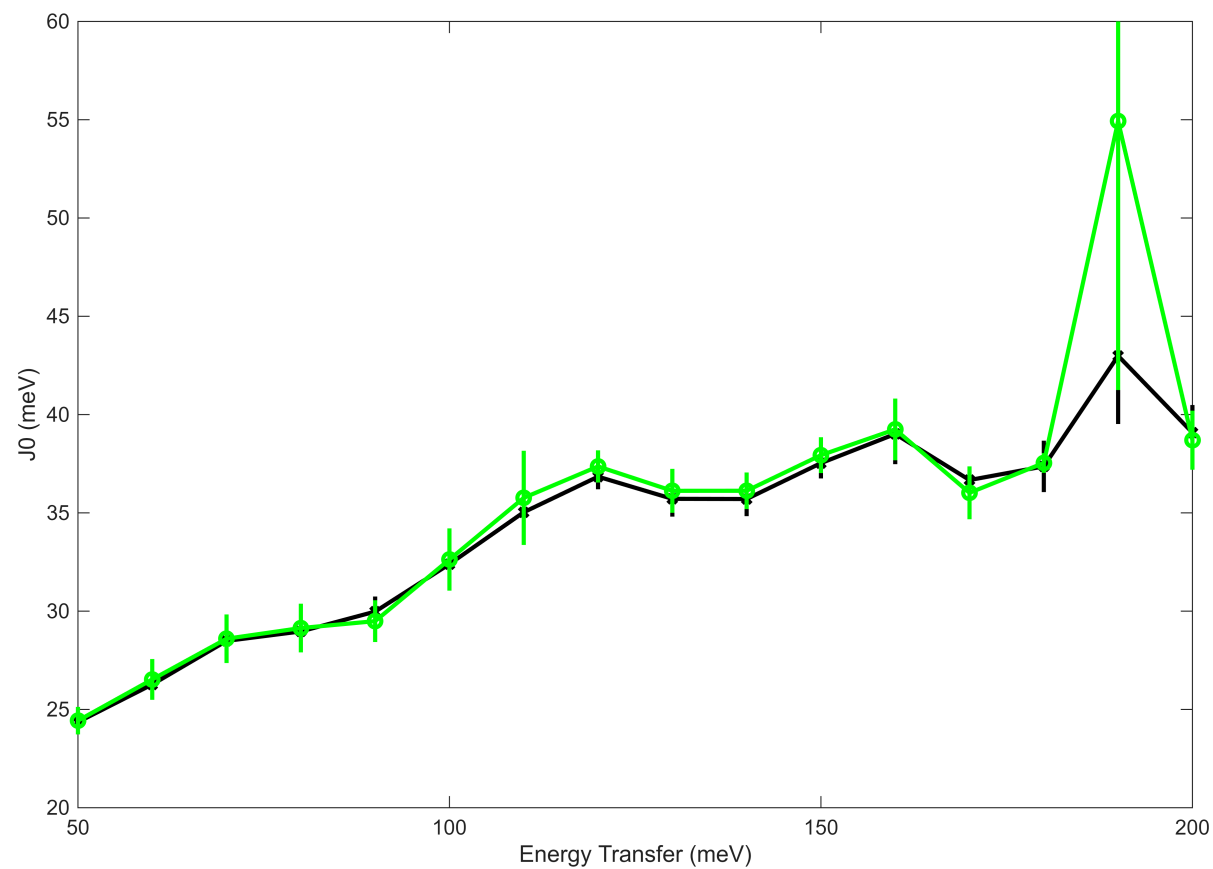
Fe_ei401_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[2+\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, \eta, 0]$, $195 \leq E \leq 205$
 $\zeta = -0.005:0.02:1.015$ in $[2, 0, \zeta]$











```

    %cl,'En_cuts800',dir_name,cut_en,dE_step,half_dE);
beep
beep
beep

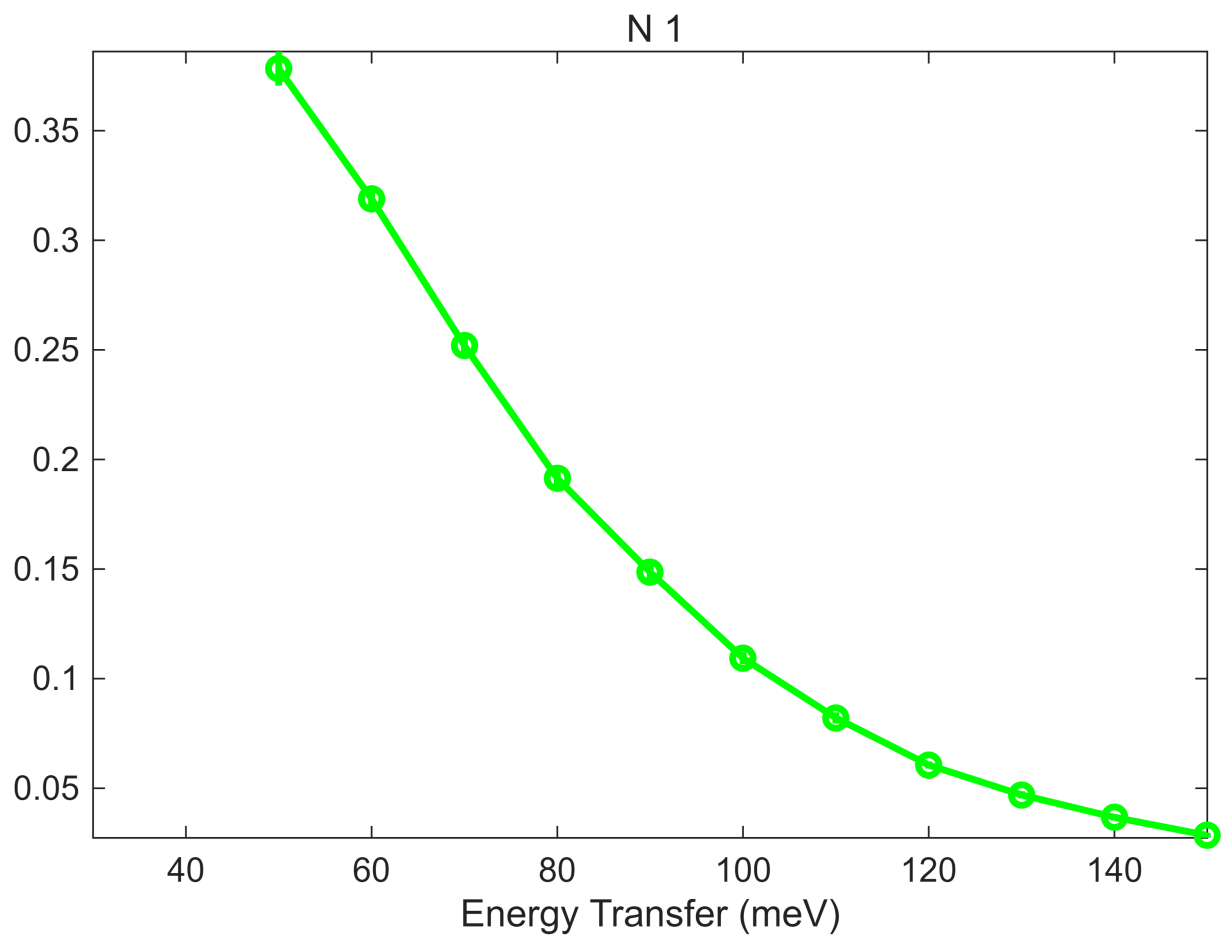
```

```

fit_res_400 = fit_res_400rec;

bg_data = extract_bg_par(fit_res_400.all_fit_par,[1,2]);
% original slope parameters
pd(bg_data(1));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_f
figure;

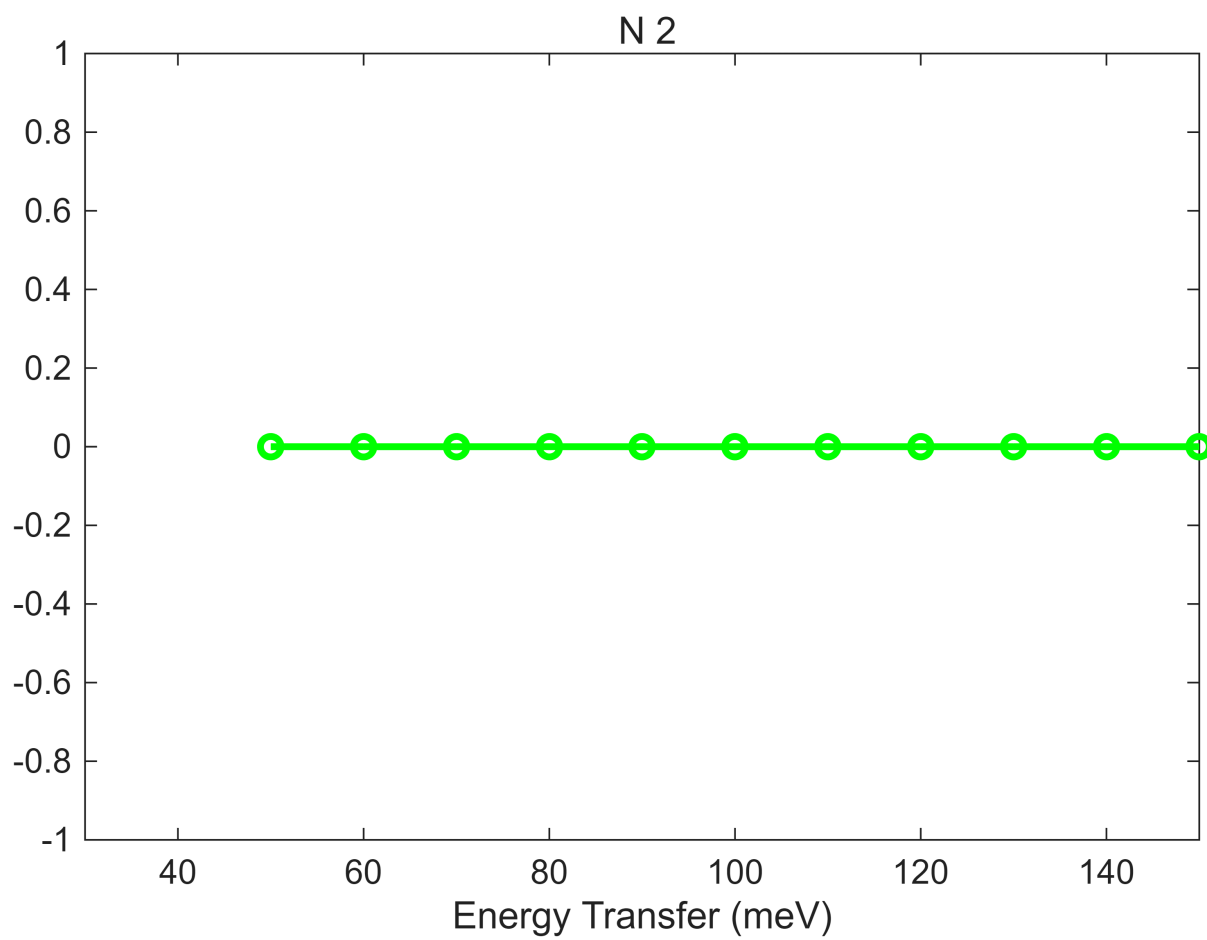
```



```

pd(bg_data(2));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_f
figure

```



```
fit_resEi400 = fit_res_400rec;  
save(res_name, "fit_resEi400")
```